

Universidad Politécnica de Madrid
Escuela Técnica Superior de Ingenieros de Caminos,
Canales y Puertos
Grado en Ingeniería de Materiales



**Evaluation of the corrosive behavior of electrodes
exposed to non-thermal plasma in atmospheric air**

Trabajo de Fin de Grado

Ángel Domínguez Sánchez

Director/es:

María Vega Aguirre Cebrián
María del Mar Cogollo de Cádiz

Madrid, 2020

Dedicado a mis abuelos, mis padres, mis tíos y mi hermano

Agradecimientos

Gracias a mi directora y tutora del trabajo de fin de grado María Vega Aguirre Cebrián por permitirme formar parte de este proyecto, por todo lo aprendido y por su gran ayuda en la parte experimental y a mi cotutora María del Mar Cogollo de Cádiz por todo lo aprendido y su gran labor en la parte de simulación.

Resumen

El fenómeno de descarga corona se basa en la generación de plasma no térmico mediante la aplicación de una elevada diferencia de potencial entre una configuración determinada por dos electrodos denominados corona y colector o masa. El proceso de ionización va a depender directamente del medio dónde se produzca esta descarga eléctrica siendo un ambiente común en aire atmosférico. Sin embargo, estas condiciones de trabajo pueden ser agresivas en periodos cortos de tiempo, afectando directamente en la vida útil de sistemas que integren este efecto. La corrosión en plasma no térmico es un aspecto fundamental para tener en cuenta en los diferentes pasos del proceso tecnológico de desarrollo de este tipo de dispositivos.

Abstract

The corona discharge phenomenon is based on the generation of non-thermal plasma through the application of a high potential difference between a certain configuration by two electrodes known as corona and collector or mass. The ionization process is directly dependent on the environment where this electric discharge is produced, being a common environment in atmospheric air. However, these working conditions may be aggressive in short periods of time, directly affecting the lifespan of systems that include this effect. The corrosion in non-thermal plasma is a fundamental aspect to take into account in the different steps of the technological process of device development.

Contents

1	Motivation and Objectives.....	1
1.1	Motivation.....	1
1.2	Objectives.....	5
2	State-of-the-Art.....	7
2.1	Corona discharge fundamentals.....	7
2.2	Polarity.....	10
2.2.1	Negative polarity.....	10
2.2.2	Positive polarity.....	11
2.3	Configuration.....	13
2.3.1	Wire-to-plane.....	15
2.3.2	TRAID.....	16
3	Introduction.....	19
3.1	Corrosion fundamentals.....	19
3.1.1	Tungsten.....	19
3.1.1.1	Reaction with air and oxygen.....	20
3.1.1.2	Reaction with water and reduction with H.....	20
3.1.1.3	Reaction with carbon.....	22
3.1.1.4	Electrochemistry of tungsten.....	23
3.1.2	Aluminum.....	25
3.1.2.1	Reactions and properties.....	25
3.1.2.2	Electrode behavior.....	26
3.2	Simulation model.....	27
3.2.1	Corona discharge model.....	27
3.2.2	Corrosion model.....	28
4	Materials and methods.....	31
4.1	Materials.....	31
4.2	Experimental and computational methods.....	33
4.2.1	Sample preparation.....	33

4.2.2	Scanning electron microscopy (SEM).....	36
4.2.3	Energy dispersive x-ray spectroscopy (EDX).....	38
4.2.4	ImageJ.....	39
4.2.5	Computational simulation.....	40
4.2.5.1	Geometrical model.....	40
4.2.5.2	Physics.....	42
4.2.5.3	Mesh and study.....	49
5	Results and discussion.....	53
5.1	Experimental results.....	53
5.1.1	Filament 1.5.....	54
5.1.1.1	Transversal image.....	54
5.1.1.2	Longitudinal image.....	55
5.1.2	Filament 2.4.....	60
5.1.3	Filament 2.8.....	62
5.2	Computational results.....	65
5.2.1	Filament 25 μm	65
5.2.2	Filament 100 μm	69
5.2.3	Erosion rate.....	71
5.3	Discussion.....	72
6	Conclusions.....	77
7	Future lines.....	79
	References.....	81

1 Motivation and Objectives

1.1 Motivation

Understanding the world around us is the biggest challenge the human being has ever faced. In order to reach this objective, the development of different tools has characterized the civilization throughout history. The technology improvement has always been tightly related to the evolution of human reasoning and intelligence. Despite the development of technology had the aim of enhance the survival capabilities of humans, nowadays, it represents, among other things, the revolution of the cognitive faculties necessary to fulfill one of the most important human features, the curiosity.

This project makes my small contribution to the science by gathering the information of the processes involved in the generation of non-thermal plasma and analyzing the corrosion undergone by the corresponding electrodes. Following, a brief explanation of the process is going to be exposed in order to make it easier to understand.

The first point is to introduce the meaning of non-thermal plasma. Plasma is a state of the matter, like solid, liquid or gas, where the particles inside are ionized or partially ionized and it has an undefined volume or shape, like the gas state. Plasma can be obtained by different processes, the most common is by heating up a gas, generating hot plasma, where a thermal equilibrium between particles exist. The non-thermal or cold plasma is characterized by maintaining the particles inside in non-equilibrium by owning, simultaneously, a really hot electron gas while the rest of the gas and ions are barely above room temperature, this is what happens in some fluorescent lamps. Many other non-thermal processes are able to induce ionization but, this project will focus on the corona discharge. This phenomenon is carried out between two electrodes, both may be composed of different

materials and shapes. Some applications of this process are ozone manufacturing, refrigeration system or inactivation of microorganism.

Corona discharge takes place at atmospheric pressure and it requires relatively low electrical power. However, the corona discharge is generated by strong electric fields associated to small diameter wires, needles or sharp edges on the electrode.

The corona discharge is a technology based on a high potential difference between two electrodes by accumulating charges inside. When the charge is high enough, the neutral air molecules surrounding the electrode will start to ionize, attracting the positive air ions immediately next to the electrode and repelling the free electrons and negative ions generated during ionization. Some of these repelled particles have enough energy to be able to ionize further atoms, however, far enough atoms will not be affected by the particles. The electric field will move ions and electrons in different directions due to their different charge, preventing them to recombine. This movement of particles within the region affected by the electric field is also called ionic wind ^[1]. During the process, the ionized atoms get excited and eventually relaxed, emitting a photon. In atmospheric air, this excitation and relaxation is carried out by elements like oxygen or nitrogen, generating a characteristic violet glow, known as non-thermal plasma; high-voltage cables are a good example where this can be observed.

The shape of both electrodes will dictate the efficiency and type of the process. There are many different efficient configurations, but this project will focus on is the wire-to-plane configuration. As it is said before, there are two electrodes performing the corona discharge. The one who perform the wire shape is called corona electrode and will accumulate charges and the other one who perform the plane shape is called mass electrode. Moreover, if the charge difference between the corona and the mass electrode, known as potential difference, is high enough, this leads to the electrical breakdown of the air, it means that the air contained between them will eventually be completely ionized, leading to a conductor path of ionized air where the charges can move through and create a spark between the corona and the mass electrode. This is called arc discharge and is the explanation given to describe lightnings bolts in nature.

Following, the materials involved are those who perform the corona and the mass electrode, different materials will undergo corona discharge and, hence, will be analyzed afterwards. There are many materials that are used as electrodes in this process, it can either

be one material or a composition of various materials, however, metals are the most common. Some of the materials used as corona electrodes are stainless steel, nickel-chromium alloys, molybdenum^[2], copper or tungsten. The mass electrode may be composed of aluminum, stainless steel or some alloys like brass, depending on the device. The main material studied in this project is the tungsten, used as corona electrode, and the material used as mass electrode will be, usually, aluminum. Both electrodes undergo more aggressive and complex corrosion than usual due to the electric current and the ion formation, but the corrosion undergone by the corona electrode is much higher. Therefore, a detailed explanation of the corrosion mechanism of the tungsten is described in the document.

Finally, some experimental techniques are used in order to evaluate the corrosion and the state of the resulting electrodes, after the corona discharge has been performed. These techniques are: Scanning Electron Microscopy (SEM), used to observe the state of the surface of the material, Energy Dispersive X-Ray spectroscopy (EDX), provides information about the composition and, finally, X-Ray Fluorescent Spectroscopy (XRF). Furthermore, a simulation by software COMSOL Multiphysics is also performed in order to make a comparison between the simulated process and the real one.

Now, in order to have an idea about the socioeconomic motivation of corona discharge throughout history, a little background is going to be exposed below.

Applications for corona discharge have existed for over a hundred years, being electrostatic precipitators for air pollution control as the main application in the beginnings. The first successful commercial electrostatic precipitator was developed by Dr. Frederick G. Cottrell in 1906 which was installed at an acid manufacturing plant^[3]. Since then, electrostatic precipitators have been widely used in industrial processes to remove both solid particles and liquid droplets from combustion sources.

Nowadays, corona discharge is extensively used in several commercial ways and is gaining attention for use in other applications, because of the corona is relatively easy to establish. Some applications are, for instance, electrostatic precipitation, electrophotography, destruction of toxic compounds, generation of ozone, ionization, static control in semiconductor manufacture and control of acid gases from combustion sources^[4].

The most ambitious application that is being pursued is the propulsion by ionic wind. Recent experiments carried out by NASA in 2009 set a thrust to power ratio of 20 N/kW and a thrust to area ratio of 20 N/m² as a goal that must be reached simultaneously so the corona

discharge might be useful ^[5]. Although the goal of 20 N/kW was reached, the thrust to area ratio was not, therefore, the use of Corona Discharge for propulsion does not seem very practical, nevertheless, there are still many characteristics to optimize.

More recently, the development of electrical devices during 21st century requires smaller and smaller chips and circuits, becoming more difficult to avoid heat flux accumulation. Corona Discharge is being investigated with the aim of creating an efficient cooling system using the ionic wind generated during the process. It is considered to have a potential to replace traditional cooling devices.

Although the corona discharge has many applications and advantages, it also has big limitations. Some of them are easy to deduce, like thrust force in the case of propulsion, however, many others are not easy to observe. To summarize, these problems are: Power, corrosion and voltage limit.

The first one to talk about is the voltage limit. This is a very important feature of corona discharge because, as it is said before, if the voltage applied in the corona electrode exceeds a limit of the potential difference between electrodes, the ionized air is going to create a conductor path between the two electrodes generating a spark, that discharges both electrodes. Hence, every time this happens, the corona discharge will lose its efficiency.

The second one is probably the most limiting one, the corrosion. Due to the ionization and the current applied, despite both electrodes degrade, the corona electrode degrades at a much higher rate than usual and higher than the mass electrode, reducing considerably the efficiency of the electrode and its lifespan. Furthermore, a ceramic oxide layer will grow on the surface of the electrode, reducing the conductive surface area and, eventually, the whole surface are decreasing the efficiency. When the time service becomes longer, corrosion, induced by the current, is able to corrode the whole cross section of the wire (corona electrode), and, sometimes, it produces the breakage of the wire, which definitely destroys the electrode.

Last but not least, the power of the corona discharge is completely limited by the other two problems, for instance, corrosion limits the power of high-voltage electrodes because of the high degradation rate and, subsequently, produces the following loss of efficiency. In addition, the power nowadays is not high enough for some applications.

As it is stated, the corrosion is a very important phenomenon that takes place during corona discharge. Therefore, it is important to understand the corrosive behavior of the tungsten and the aluminum. In addition, the simulation by COMSOL is of great importance to obtain comparative results. Hence, these fundamentals are exposed in chapter three.

1.2 Objectives

Once we know the main problems of corona discharge, it is easy to conceive that the study of the degradation of electrodes is really important. The prior aim of this project is to identify the corrosion processes involved in tungsten corona electrodes by studying the tungsten corrosion mechanisms, microstructure and composition in the corroded state, as well as consider different materials that could perform as corona electrode, with the objective of enhance the lifespan and time service of these electrodes as the maximum voltage and maximum power during the process. But Covid19 circumstances have forced a work with results and resources available during confinement. Therefore, the objectives have been as following:

- The study, by means of SEM and EDX, of different filaments under different work condition of corona discharge, in order to determine the corrosion produced in these conditions.
- To develop a computational model simulated by COMSOL, with the aim of correlate the corrosion of the electrode and conditions of corona discharge.

2 State-of-the-art

2.1 Corona discharge fundamentals

First of all, a detailed explanation of the fundamentals of the corona discharge processes are going to be provided in this subpart in order to understand all the mechanisms behind these processes and the subsequent effects on the materials involved and the air surrounding the devices where these processes take place.

The electrical fundamentals behind the corona discharge are not different from other processes previously discovered. As we know, the corona discharge consists in two electrodes that generate a potential difference between them by charging them up, it can be either one of them or both. The charge accumulation on the surface of one or both electrodes is produced as a result of an electric current generated through the electrode. In addition, the electric current induces an electric field around the electrode which can be easily related with particles features by two basic formulas in physics:

$$F = qE \quad (1)$$

$$F = ma \quad (2)$$

With the charge of the particle (q), electric field (E), mass of the particle (m), acceleration (a) and force (F), it is possible to theoretically calculate the acceleration undergone by a certain particle at a given electric field. These formulas consider a constant electric field along the region and are standard for point charges, in addition, further considerations like interactions between particles are not taken into account. It is easy to conclude that the acceleration of a certain particle depends on the electric field that surrounds the particle and interact with. The expression of the electric field is different for every certain

geometry; therefore, the electric field is quite important and minimum variations in the electrode lead to big changes in the value.

The electric field is very important in electronic technology and, in atomic scale, it is responsible of the attractive forces between nucleus and electrons. Regarding corona discharge, this magnitude is tightly related to the voltage and the geometry of electrodes, where both of them are necessary to confine the reaction in a region so it prevents from spreading in all directions. Because of the corona electrodes studied in this project are wire-shaped, following there is a description of its electric field effects.

One of the most characteristic features of the wire corona electrode is its curvature, making corona discharge more efficient when the curvature is sharper, at a constant voltage. In order to understand this phenomenon, basic fundamentals of electric wires are explained. First of all, it is observed in (3) that the electrical field generated by a straight wire at a specific point in the space is:

$$E = \frac{\lambda}{2\pi\epsilon R} \quad (3)$$

Where it is correlated the charge density of the wire (λ), the electric permittivity of the surrounding substance (ϵ) and the distance between the wire and the point to calculate (R). This formula gives a value of electric field for an ideal wire, this means that the value to be calculated is the upper limit of the electric field. As it can be deduced, the charge density of the wire (λ) depends on the voltage applied. However, this formula is incomplete for two main reasons, because it takes into account, among other things, that the electric field decrease rate is constant with the increase of R , but, the most important thing, is that it assumes that the electric field tends to infinite when R tends to be zero, in other words, the electric field in the surface of the wire is infinite, according to this formula.

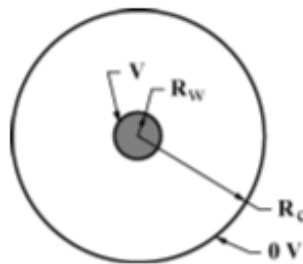


Figure 1^[9]. Visual representation of equation (4).

In order to be a little more specific with the surface of the wire, there is a good approximation in equation (4)^[9], that is represented in Figure 8, and it is observed that the electric field at the wire surface depends on the voltage applied, the outer cylinder radius (Rc) and the wire radius (Rw):

$$E = \frac{V}{\left[R_w \ln \left(\frac{R_c}{R_w} \right) \right]} \quad (4)$$

Following, an approximation of the electric field on the surface of the wire can be obtained. Looking at the equation (4)^[9], it is easy to deduce that at a constant V and Rc, the value of E is dependent of the wire radius Rw. Finally, if the value of Rw decreases, the value of the natural logarithm increases, however, the decrease rate of Rw is higher than the increase rate of the natural logarithm. To conclude, it is proved that the decrease of the wire radius Rw increase the electric field on the surface of itself, so the corona discharge is more efficient at the same voltage and Rc.

As it is stated in the introduction, the corona discharge processes are involved in the generation of non-thermal plasma (NTP) because of the way it works. As the potential difference between both electrodes is raised, the gas around the more sharply curved electrode (corona electrode) breaks down at a voltage less than the spark breakdown, phenomenon that discharges the electrodes. This incomplete breakdown appears as a high active region of glow around the corona electrode^[3], as it is represented in Figure 1. The electrical breakdown of air molecules is known as ionization, where an insulator becomes partially conductor. The excitation and relaxation of the electrons around the nucleus of air molecules when ionized, produces the glow, also called cold or non-thermal plasma. The NTP contains a substantial difference in temperature between electrons and particles. While particles are close to room temperature, electrons are extremely above, reaching temperatures typically in ranges of 10 000 K to 250 000 K^[6]. This high-energetic mass of electrons is able to easily induce further reactions, leading NTP a remarkable ability to generate chemical reactions at atmospheric pressure and room temperature which enables a rapid start-up and shut-down of the process, providing an easy control.

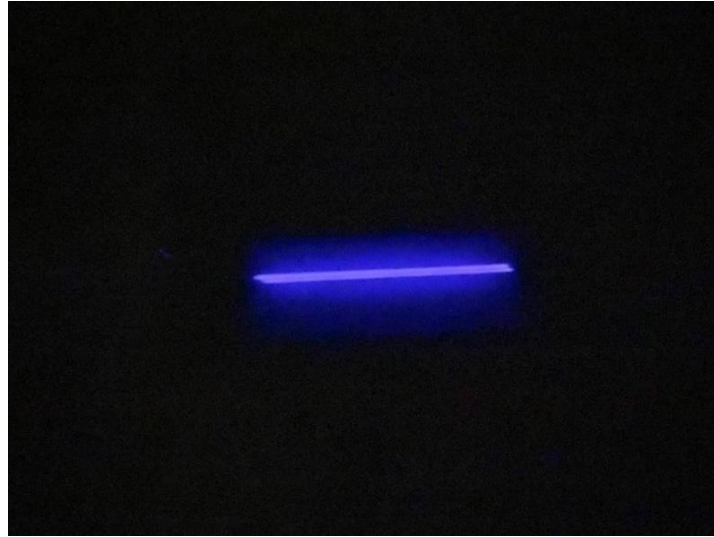


Figure 2. Violet non-thermal plasma glow produced by corona discharge performed in a wire-shaped tungsten electrode.

The air molecules and electrons that are affected by the electric field produced by the corona electrode have essentially two interactions: they are either repelled or attracted to the electrode. These two different movements through the region depend on a characteristic feature of electric charges and, hence, corona discharge: the polarity.

2.2 Polarity

An important concept in corona discharge is the polarity of the electrodes that generates the electric field, as it is said before. The repulsion or the attraction of particles is dictated by the polarity. In general, selection of polarity depends on a large number of factors, which include electrode material, device geometry, ozone generation constraints, fluid composition and others ^[8]. Despite there are many types of polarity like negative, positive, bipolar, AC or HF, this project will focus on the most important ones, which are the negative and the positive.

2.2.1 Negative polarity

The negative polarity is generated by a negative electric current moving through the corona electrode. This current produces a negative charge accumulation on the surface of the electrode, producing an electric field which attracts positive ions towards the surface of the electrode and repels free electrons and negative ions away from the electrode, generating

an electron avalanche, as observed in Figure 2, that is collected by a less curved electrode. However, unlike the positive corona process, the secondary electrons in a negative corona are produced primarily by photo-emission from the corona electrode surface. Material selection has an impact on negative corona discharge, because the production of electrons by photo-emission partially depends on the matching energy between the photons and the work function of the corona electrode material^[8]. As shown in Figure 2, the ionization region is a domain where the electric field is critical and induce ionization. In addition, its threshold is produced when the rate of ionization of molecules equals the recombination of electrons and ions. Unlike positive polarity, electrons are repelled from the corona electrode, leading to further reactions even though outside of the ionization region, this region where reactions can occur is called plasma region, visualized in Figure 2.

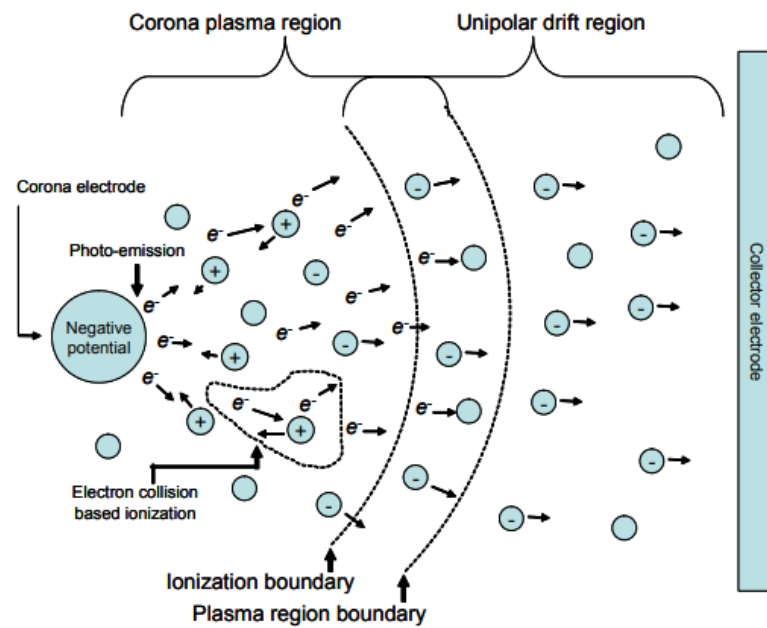


Figure 3^[8]. Model of a negative corona electrode in a wire-to-plane configuration.

2.2.2 Positive polarity

Conversely, a high positive electric potential is applied to the high tip curvature corona electrode, where electrons and negative ions will move towards the electrode, causing the breakdown of gas molecules, and positive ions are repelled away from the corona electrode in the form of a corona glow. Nevertheless, negative ions are not able to reach the electrode surface because of its instability. In the ionization region, the collision between

free natural electrons and neutral air molecules results in the release of electrons from the neutral molecules, producing ionization and, therefore, electron avalanche. Furthermore, photo-emission also takes place in positive corona, increasing the electron avalanche. As it is said before, the location where the ionization rate equals the recombination rate is the ionization and plasma region threshold, shown in Figure 3, and it is characterized for being uniform along the electrode surface that faces the collector or mass electrode. One has the choice to generate positive or negative polarity, this project focus on the positive corona, moreover, different studies have proved that each one has different properties from each other. For instance, in the introduction, it is mentioned that the corona discharge is a process utilized in ozone production. A study carried out in 2004 by Hyun-Ha Kim proved that a negative corona produces up to 7 times more ozone than a positive one^[6], in addition, ozone production is 10 times less efficient in positive corona than negative^[9]. Other recent studies state that negative corona yields a higher current than positive at a given voltage^[3], where we can conclude that negative corona corrodes at a higher rate than positive, which is really interesting when the goal depends on the movement of particles and not on the reactions, such as cooling devices.

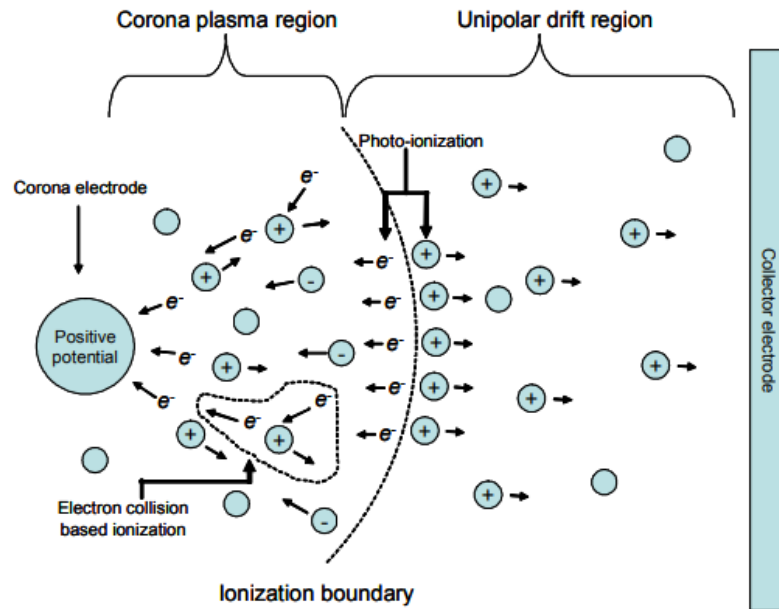


Figure 4^[8]. Model of a positive corona electrode in a wire-to-plane configuration.

Corona discharge is, by definition, a discharge where the geometry confines the gas ionizing processes to high-field ionization regions around the active electrodes^[7]. Because

of this, corona discharge exists in many different types depending on polarity, previously explained, and the configuration of the electrodes, in other words, their geometry.

2.3 Configuration

The configuration is the geometry of the electrodes used to perform a corona discharge process. The geometry has a high repercussion in the efficiency of the process and, according to different studies, the geometry does not determine the polarity of the discharge, it is possible to have positive and negative polarities using the same configuration^[4]. There are many different configurations of electrodes but the most relevant are wire-to-plane, wire-to-cylinder, needle-to-plane, needle-to-ring, wire/needle-to-grid and TRAID^[1], which is an innovative configuration for electrohydrodynamic cooling systems based on the patent by CEDRION C.T.I. S.L. It is observed that various configurations are suitable to perform corona discharge. Although an introduction to some of the configurations is provided below, this project will focus on wire-to-plane, primarily, and TRAID, therefore, they will be the last ones to be explained.

In order to know the similarities and differences, two configurations are compared. In needle-to-plane configuration, the corona electrode is a needle-shaped material whereas the mass or collector electrode is plane, represented in Figure 4. There exist substantial differences between this configuration and others, apart from geometry. In Figure 4 it is observed that the positive and negative processes of corona discharge are different. It is important to remark that Figure 4 describes all the steps until the electrical breakdown of the air between the electrodes, which needs to be avoided. Some studies have proved that positive corona discharge generates a glow that spreads over the point tip and, for hemispherical heads of more than 2 mm in diameter, eventually, it divides into two opposite alternately-emitting glows, as observed in Figure 5, because of large current values increases the frequency of relaxation and oscillation leading to lateral spread of the discharge due to the clearing time of space charge is too different between center and periphery^[10]. As we know, current increases when curvature increases at a constant voltage, however, many authors claim the needle-to-plane configuration is estimated to have a limit of 50 μm of point radius^[11]. Lower radius values do not induce a current increase.

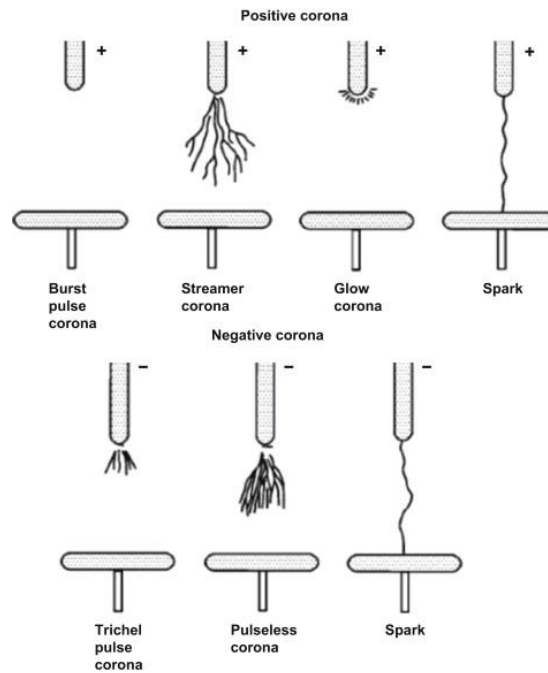


Figure 5^[4]. Positive and negative corona discharge processes in a needle-to-plane configuration.

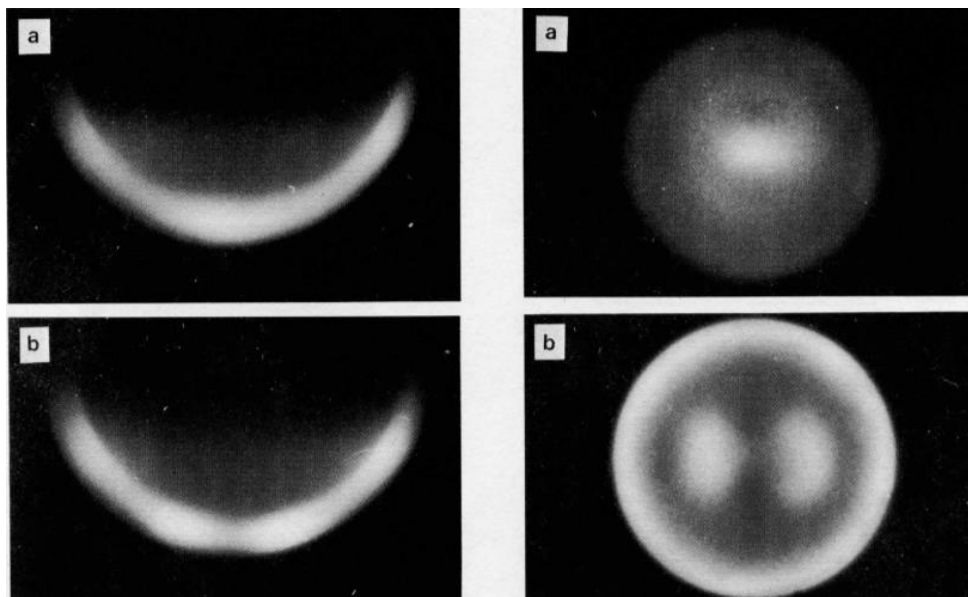


Figure 6^[10]. Positive glow of corona discharge in a needle-to-plane configuration in uniform distribution of glow (a) and lateral spread of glow (b). Lateral and front perspectives.

The main difference between cylinder-to-plane configuration and others is that both electrodes have a significant curvature, being the corona electrode curvature lower than the mass electrode. Some studies have been performed in order to find out the viability of this configuration, being the one carried out at Instituto Nacional de Técnica Aeroespacial (INTA) one of the most recent ones, specifically in 2019^[11]. In this study, a test of a device represented in Figure 6 was subjected to positive and negative corona discharge.

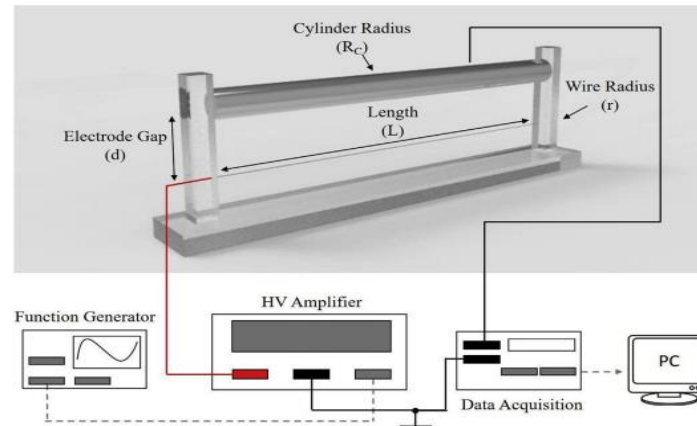


Figure 7^[11]. Arrangement of a wire-to-cylinder configuration testing.

This study concludes that the decrease of the wire-shaped corona electrode radius, unlike needle geometries, does not have a visible limit of current increase at a constant voltage^[11], in other words, the sharper is the curvature of the wire-shaped corona electrode, the stronger the electric field generated. This is very interesting when a low voltage to perform corona discharge is required, allowing us to increase efficiency maintaining voltage under the limit.

2.3.1 Wire-to-plane

The wire-to-plane configuration is the one this project will focus on. This configuration, as it is deduced from its name, is composed of a wire performing as corona electrode and a plate performing as mass or collector electrode, as observed in Figure 9. As the previous wire-to-cylinder configuration, one of the advantages is that if the curvature is sharper, its current and electric field at a constant volume is higher too. This configuration has many similarities with TRAID but the main difference is that the mass electrode is flat and it does not count with dielectric barriers, however, those barriers can be added as well.

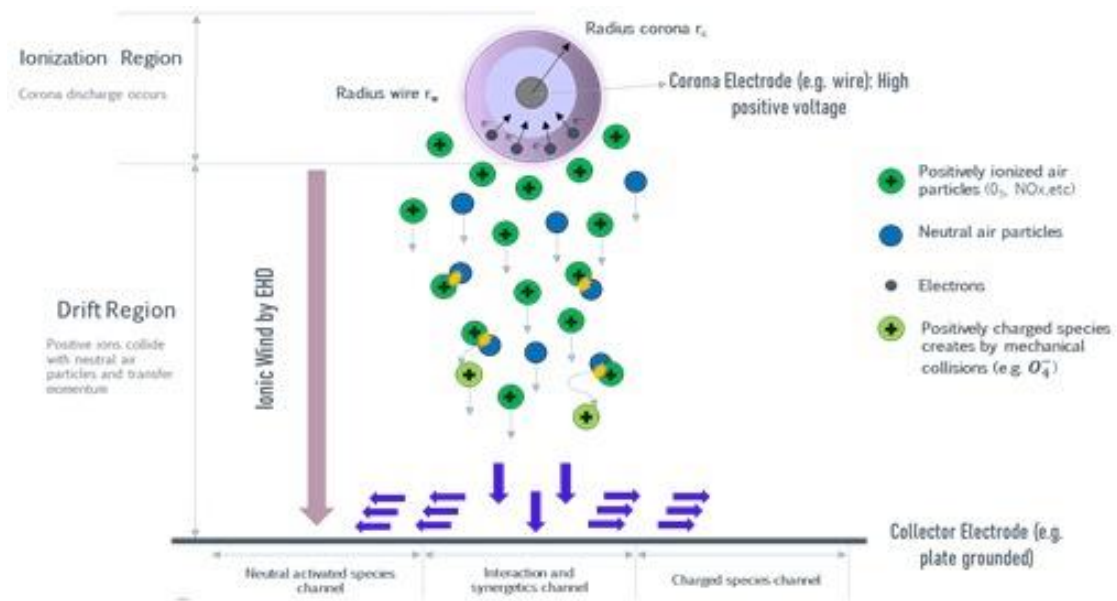


Figure 8. Positive corona discharge in a wire-to-plane configuration.

2.3.2 TRAID^[1]

This innovative configuration based on the patent by CEDRION C.T.I. S.L. is studied for electrohydrodynamic (EHD) cooling devices and is represented in Figure 7. It is observed that the configuration is based on a micrometric wire as emitter or corona electrode and a concave-shaped collector or mass electrode, the configuration has been designed from wire-to-plane. In addition, dielectric barriers have been integrated into the sides of the wire in order to separate the inlet and outlet of the fluid. The high efficiency of TRAID devices is remarkable, even though power consumption in a corona refrigeration device is around twice as high in the studied sizes, savings in mass and volume are a key parameter in many applications such as aeronautics, where the powers of those devices are negligible. Another good point, is that the electric arcs generated during the corona discharge process will remain inside due to the geometry of the mass electrode and the dielectric barriers, increasing the safety of the device.

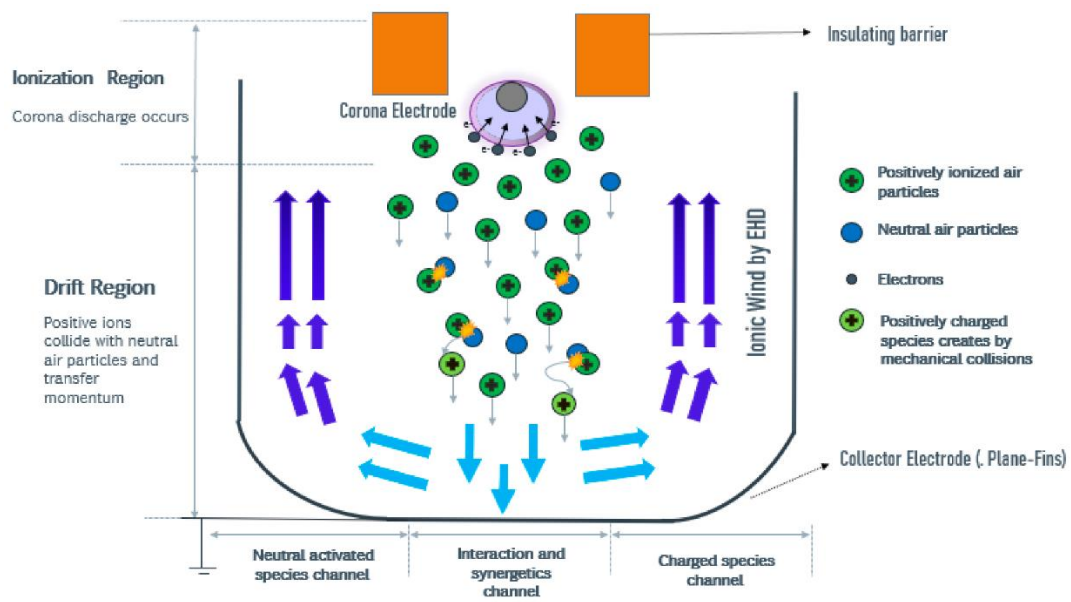


Figure 9^[1]. Positive corona discharge in a TRAID configuration.

3 Introduction

3.1 Corrosion fundamentals

The corrosion fundamentals regard the corrosion of the materials who perform as corona electrode and the mass electrode. The electrodes are surrounded by air, in addition, corona electrode generates a plasma that makes the environment around it even more aggressive. Furthermore, both electrodes are subjected to erosion caused by the movement of particles produced by corona discharge. The corrosion mechanisms of the corona electrode are, in general, the mechanisms that dictates the viability of the device, because is the one that corrodes faster, heavily reducing the efficiency of the device and, eventually, breaking the filament. Therefore, this section is going to focus in tungsten and aluminum corrosion mechanisms, because they are the main materials used in this project as corona electrode and mass electrode respectively.

3.1.1 Tungsten ^[12]

In order to make a general brief introduction of tungsten a material, some properties are exposed following. Tungsten is a metallic transition material categorized as a refractory material with the highest melting point among all elements, $T_m = 3422^\circ\text{C}$. It is a high-density material being this value of $\rho = 19.35 \text{ g/cm}^3$, which is more than twice the density of steel. This is a scarce material in the Earth but is commonly found in the form of salts and oxides. It is a material with many different applications such as nickel superalloys, as an encouraging element, but the most widely known is the application as light bulbs filaments.

The processes studied in this section are the ones that may occur between tungsten and the environment during corona discharge processes. In this case, air is the surrounding substance, therefore, the main reactions between tungsten and air compounds are studied. In

addition, the electrochemistry of tungsten is also studied in order to understand its properties and its behavior when is acting as a cathode or/and anode.

3.1.1.1 Oxidation of tungsten by air or oxygen

Tungsten has a high sensitivity toward oxygen that restricts its high-temperature applications to vacuum or protecting atmospheres only^[12]. Oxygen is adsorbed on tungsten surfaces in a variety of atomic and molecular states. At low temperature ($<0^{\circ}\text{C}$), oxygen is adsorbed molecularly, but at room temperature, a covered surface shows an ordered oxygen superstructure. If the temperature is increased, oxide structures are formed, generating a surface layer. Following, oxygen diffuses from the surface into the crystal lattice forming a bcc (α) solid solution, however, the solubility is very low^[12]. Once the oxygen has been adsorbed and diffused through the material, the oxidation starts.

The oxidation of tungsten starts at room temperature and depends strongly on temperature and on the oxygen partial pressure at elevated temperatures. The oxide formed in oxygen as well as humid or dry air is always tungsten trioxide (WO_3). The thickness of the oxide layer increases slowly with temperature, but it increases faster above 200°C . From 327 to 400°C a protective layer is formed, but, above 500°C this layer cracks. An important process is that at 750°C the sublimation of WO_3 starts and increasing with temperature, becoming the rate of sublimation equal to the rate of oxidation^[12], hence, the surface is oxide free because as soon as it oxidizes, it sublimates. In addition, some recent studies state that WO_3 is a semiconductor that has excellent sensitivity and selectivity on ozone and it is used in ozone monitoring^[13].

Finally, above 2100 K gaseous WO_2 is formed more rapidly than WO_3 and, above 2500 K the volatilization of WO_2 occurs.

3.1.1.2 Reaction of tungsten with water and reduction with hydrogen

Because of water exists dissolved in the air in the form of humidity, it is important to study the possible reactions of tungsten with water. The basic equation for the reactions involved is:



Equation (5) is a general description of the main reactions where the oxidation processes involve this equation from left to right, whereas the reduction processes involve this equation from right to left.

Water reacts with tungsten below its boiling point. In principle, water acts as an oxidizing agent. According to some investigations WO_2 , WO_3 and oxide hydrates are usually formed^[12]:



The reaction with distilled water is very slow at low temperature: the corrosion rate is $3.8\mu\text{gW/m}^2\text{h}$ at 38°C ^[12]. Following, the formation of a non-protective oxide layer is carried out. Corrosion rate increases with temperature and pressure.

Reaction with water vapor or humid air produces a WO_3 layer a temperature range of $20\text{-}200^\circ\text{C}$, right the kind of temperatures at which corona discharge is performed (room temperature generally). Furthermore, the thickness of the oxide layer increases when humidity is increased. The reaction rate depends on the temperature as well as on the partial pressure of H_2 and H_2O . Water molecules are adsorbed on the surface of tungsten and dissociates, then the oxygen atoms diffuse through the material forming at first a solid solution and, subsequently, an oxide compound while hydrogen escapes.

Depending on temperature and p_{H_2O}/p_{H_2} all known oxides can be formed. At low values of temperature and coefficient, the reaction product is WO_2 . However, an increase of both parameters yields in the formation of higher oxides like $WO_{2.72}$, $WO_{2.9}$ and WO_3 . Knowing these possible compounds, water vapor may be responsible of, among other things, the destruction of tungsten in contact with alumina (water vapor leads to WO_2 formation that reacts with Al_2O_3 to form $AlWO_4$).

Like tungsten is usually in the form of oxides in nature, one of the best methods to obtain tungsten is by reducing tungsten oxides with hydrogen according to the following equation:



At temperatures between 600-1100°C in a streaming hydrogen atmosphere. The reduction process is not only a chemical conversion from oxide to metal, it is rather a complex process where some impurities can interfere in the reactions. Although it is an important reaction, as it is stated before, it only takes place at relatively high temperatures.

3.1.1.3 Reaction of tungsten with carbon or carbon-containing compounds

Carbon is a common element that can be found in air usually in the form of carbon compounds, specially oxides. Therefore, the possible reactions of tungsten with carbon have to be studied in order to know the effects of carbon on the material and the most common compounds that are formed as a consequence of the interaction between carbon and tungsten.

This section deals with the carburization of tungsten and the most common process to produce WC. Tungsten reacts with solid carbon as well as with a variety of carbon-containing compounds to form carbides, generally, at high temperatures. Nevertheless, it has been observed that solid carbon layers may react with tungsten layers to form W₂C at room temperature, and, WC at 800°C.

Regarding the properties of tungsten, polycrystalline metal carburizes more easily than monocrystalline metal, this is because lattice defects strongly increases the diffusivity of carbon in tungsten. Furthermore, the purity of tungsten has an influence on the carburization. Certain impurities, such as cobalt, accelerate the carburization and others, like alkali metals, retard it^[12].

In addition, the reduction of tungsten oxides by carbon or carbon-containing compounds may occur. The reduction can be easily performed, however, statements about the starting temperature for the reaction between WO₃ and solid carbon vary in the current literature between 655°C and 783°C. Differences in WO₃ and C properties as well as atmospheres may be responsible of that. The temperature ranges are high, for instance, carbon monoxide starts to react with WO₃ at 535°C.

To conclude, carbon can only be found in the tungsten electrode either as an interstitial element or in the form of W₂C because of the corona discharge is performed at

room temperature and those are the only possible reactions that are able to occur at this temperature.

3.1.1.4 Electrochemistry of tungsten

This is one of the most important sections regarding tungsten, because of corona discharge is based on a potential difference between two electrodes and the environment contains water in the form of humidity, electrochemical processes are involved.

The electrochemical behavior of tungsten in aqueous solutions is very closely related to the high affinity of tungsten toward oxygen and the complexity of tungsten species in aqueous solution. Tungsten, due to its affinity toward oxygen, may never form a metal/metal ion electrode system, otherwise, it will form a metal/metal oxide/metal ion system, which is quite irreversible. The oxide coating of the metallic tungsten is strongly influenced by the ions and equilibria of the solution, which are closely related to pH, temperature, concentration, presence of agents, etc. Therefore, not only the chemistry of tungsten but the oxides formed are complex and not always understood. In order to introduce the general idea about the corrosion of tungsten, it is necessary to talk about its Pourbaix diagram, represented in Figure 10. In this diagram it is observed the regions of pH and potential where the tungsten each one of the 3 different states. The immunity region is the area where the material does not undergo corrosion, the passivation region is the area where the material develops a passive layer that prevents itself from corrosion and, finally, the corrosion region is the area where the material corrodes.

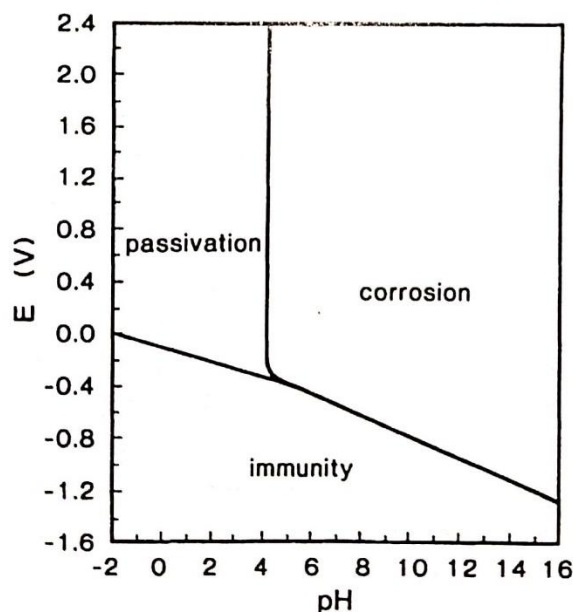
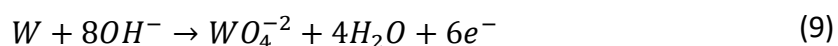


Figure 10^[12]. Pourbaix diagram of tungsten

The cathode reactions in tungsten form immediately an oxide layer, depositing high molecular oxides. WO_3 is a n-type semiconductor^[14] and is chemically reduced at the cathode. Therefore, a high potential in the interface is initiated and, hence, the reduction. There is an increase of conductivity in WO_3 generated by the changes in the electronic structure of W atoms.

The anodic reactions in basic solutions lead to an electrolytical dissolution of tungsten which is a complex process that can be summarized in the following equation:



As it is described before, positive corona discharge lead to a positive charge accumulation on the surface of the material, that eventually ionizes the surrounding air and produces a repulsion of positive ions and an attraction of negative ions and free electrons. Those particles produce further reactions with molecules generating what is known as electron avalanche. Some different studies^{[15][16]} proved that the electron attraction produce impacts on the surface of the corona electrode and it induces erosion in the material. This erosion produces the release of tungsten and oxides previously formed into the air producing a mass loss. Furthermore, the aggressive environment where the tungsten is subjected to corona discharge, is even more aggressive because of the erosion, so it is not rare to observe the breakage of the thin filament. Some studies estate that tungsten is one of the most susceptible materials to erosion^[15], being more susceptible than stainless steel or platinum,

for example. It has to be taken into account that the species released into the air are also able to move in different directions, producing an impact not only in the corona electrode but the mass electrode. The most common specie is WO_3 and some others may also be formed, such as $(\text{WO}_4)^{-2}$ due to the presence of humidity ^[16].

3.1.2 Aluminum

This material is a metallic element characterized for being light and abundant in our planet. Because of this, it has many uses, specially, in aeronautic industry in the form of alloys. Despite it is a light, ductile and resistant metal, it is not recommendable its utilization at service temperatures above 200°C ^[17]. Although it is a very reactive metal, aluminum has a good corrosion resistance due to the properties of its passive layer. In this section the possible reactions and processes during corona discharge are going to be studied.

3.1.2.1 Reactions and properties

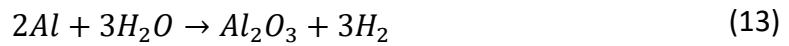
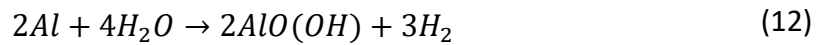
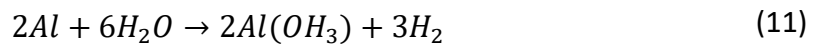
Aluminum corrosion resistance is highly dependent on the alloy elements, that can vary its resistance making it better and, sometimes, worse. Pure aluminum is more corrosion resistant than any other aluminum alloys ^[18] and it does not incur any of the more drastic forms of localized corrosion such as stress-corrosion cracking (SCC) or intergranular corrosion.

Uniform or general corrosion of aluminum is not common, except in highly acid or alkaline corrodents, in special. However, if the oxide layer is dissolved in the environment or destroyed as a consequence of ion impacts, it may corrode uniformly at a steady rate. The main oxide formed due to the reaction between aluminum and air is Al_2O_3 , also known as alumina, according to the following reaction:



Alumina is a stable oxide until high temperatures, reaching temperatures of 2000°C , becoming a very important oxide layer in high temperature alloys such as nickel-based superalloys ^[19].

One of the most important corrosion mechanisms of aluminum is pitting. This mechanism takes place at localized discontinuities in the oxide film on aluminum exposed to atmosphere, fresh or salt water, or other neutral electrolytes. Pitting occurs in a range of pH from 4.5 to 9^[18]. The pits can be small and concentrated or they can be widely scattered and with different sizes, depending on the type of aluminum alloy, oxide film or nature of the corrodent. The chloride ion is known to facilitate the breakage of the oxide layer, generating pitting corrosion. Because the environment lacks on chlore, no pitting will be observed, initially. Then, the reaction of the aluminum with the humidity may be one of the following equations:



All three compounds can be formed, however, between room temperature and 280°C the most stable one is Al(OH₃), hence, it will be the main product from the reaction between aluminum and water.

3.1.2.2 Electrode behavior

As it is said before, aluminum is the main material used as collector or mass electrode during the corona discharge processes studied in this project. The situation of this electrode is different from the corona electrode. Despite it does not have a current through the material, it receives all the positive ions repelled from the corona electrode, generating a more aggressive erosion that, combined with the aggressive environment, produces a great increase in the corrosion rate compared to aluminum not subjected to corona discharge. It must be taken into account that the ions colliding with the mass electrode are positive and they are chemically active species, this means that are more capable to produce oxidation on the contact surface. In addition, the movement of the oxidizing agent maintain the corrosion rate so it cannot decrease.

3.2 Simulation model

In order to perform a simulation of the corrosion of the electrodes, it is necessary to introduce the equations that are going to be involved and the simulation of the program are based on. As it is said in the introduction, the program used to perform the simulation is COMSOL Multiphysics. In this section, the numerical equations of the phenomena of corona discharge and corrosion are going to be exposed.

3.2.1 Corona discharge model

First of all, the computational model of corona discharge is based on the electrostatic forces formed during the process and the charge and particle movement through the environment, in this case, atmospheric air. The electrostatic process must fulfill the electric charge conservation equations^[20]:

$$\nabla D = \rho_v \quad (14)$$

$$E = -\nabla V \quad (15)$$

$$\nabla(\varepsilon_0 \varepsilon_r E) = \rho_v \quad (16)$$

Where D stands for the ion diffusion coefficient, E is the electric field, ∇ corresponds to the nabla operator, V is the electric potential, ε_0 represents the electric permittivity in vacuum, ε_r represents the relative permittivity of the material or environment and, finally, ρ_v is the ionic space charge density. Both equations, (15) and (16), can be merged to form the Poisson's equation, that govern the electric field:

$$\nabla^2 V = -\frac{\rho_v}{\varepsilon} \quad (17)$$

Here, the electric field intensity is given as a gradient of the electric potential, thus satisfying the Poisson's equation. Because of the electric potential produces a charge accumulation on the surface, this process also is considered by the program, represented in the following equations:

$$\frac{\partial \rho_s}{\partial t} = nJ_i + nJ_e \quad (18)$$

$$-n(D_1 - D_2) = \rho_s \quad (19)$$

Where J_i and J_e are the ion flux and electron flux respectively, n is the normal vector and D_1 and D_2 are the electrical displacement vector in the material and air, respectively^[20]. Furthermore, the ionic charges must satisfy the charge transport equations as well as the Poisson's equation. These equations are the following^[20]:

$$\nabla J = S \quad (20)$$

$$J = z_q \mu_i \rho E + \rho u \quad (21)$$

Where J represents the flux, S is the current source, z_q is the charge number, μ_i corresponds to ion mobility and u is the gas velocity field. In addition, the term ρu only appears when the transport is carried out by convection, which is the case. Doing the same as with the charge conservation equations, equations (20) and (21) are merged, and applying equation (14), the global charge transport equation is obtained:

$$S + \nabla^2 D(u - \mu_c E) = 0 \quad (22)$$

Once the equations of the corona discharge module implemented in COMSOL are stated, now it is time to introduce the equations of the corrosion module.

3.2.2 Corrosion model

Corrosion module allow us to choose different corrosive situations, such as primary corrosion, secondary corrosion and tertiary corrosion Nernst-Planck. Primary and secondary describe the current distribution, potential distribution and geometrical changes in a corrosion cell under the assumption that the variations in composition in the electrolyte are negligible. Furthermore, the primary corrosion considers that the activation overpotentials of the electrode reactions are negligibly small, whereas, secondary corrosion does not consider small activation overpotentials.

Tertiary corrosion Nernst-Planck takes into account the transport of charged and neutral species in the electrolyte described by diffusion, migration, and convection. The three of them combine current distribution and deformed geometry interfaces. In addition, tertiary corrosion also considers Nernst-Planck interface.

The first equation to be introduced is the Nernst-Planck equation, that determines the flux in the electrolyte by adding the diffusion, convection and migration phenomena^[21]. This equation is described below:

$$N_i = -D_i \nabla c_i + c_i u - z_i m_i F c_i \nabla \phi_l \quad (23)$$

Where D_i is the diffusivity, c_i the concentration, u corresponds to the flow velocity, z_i is the charge, m_i is mobility, F is the Faraday constant and, finally, ϕ_l is the electrolyte potential. This equation leads us to equation (24) that, developed it is converted in equation (25). These equations are described following:

$$j = F \sum_i z_i N_i \quad (24)$$

$$j = F \left(\sum_i -z_i D_i \nabla c_i + u \sum_i z_i c_i - \nabla \phi_l \sum_i (z_i)^2 m_i F c_i \right) \quad (25)$$

From equation (25), some conditions may be applied in order to simplify the expressions. Electroneutrality condition considers charge conservation, therefore, the sum of charges is equal to zero. Resulting in the following equation:

$$j = F \left(\sum_i -z_i D_i \nabla c_i - \nabla \phi_l \sum_i (z_i)^2 m_i F c_i \right) \quad (26)$$

If perfectly mixed primary and secondary current distribution applies, it results in equation (27), described below:

$$j = -\nabla \phi_l \left(F \sum_i (z_i)^2 m_i F c_i \right) \quad (27)$$

With the term between parenthesis being described as conductivity and represented with symbol kappa (κ).

Finally, the conservation equations are introduced following. These kinds of equations are the ones that describe the distribution and the conservation of current or species. In the following equation the conservation of species is described:

$$\frac{\partial c_i}{\partial t} = -\nabla(-D_i \nabla c_i + c_i u - z_i m_i F c_i \nabla \phi_l) + R_i \quad (28)$$

Where R_i is the reaction rate. Equation (28) describes conservation of species under electroneutrality condition. Moreover, the conservation of current, Nernst-Planck with electroneutrality is described following:

$$\nabla \left(F \left(\sum_i -z_i D_i \nabla c_i - \nabla \phi_l \sum_i (z_i)^2 m_i F c_i \right) \right) = 0 \quad (29)$$

This equation can be simplified to primary and secondary current distribution leading to the last equation:

$$j = \nabla(-\kappa \nabla \phi_l) \quad (30)$$

Now that all the main equations regarding corrosion module have been introduced, the difference between the types of current has to be emphasized in order to know what equations are involved. Primary current distribution only assumes ohmic potential drops^[21], there is no current/potential contribution from delayed kinetics (all reactions on the surfaces are simultaneous). Secondary current adds the effects of activation overpotential^[21] (Butler-Volmer kinetics or similar). Finally, the tertiary current adds mass transport effects.

Once the theoretical fundamentals of corona discharge, corrosion and the mathematical model have been exposed, the materials and methods utilized to characterize and analyze the electrodes are detailed in the following chapter.

4 Materials and methods

4.1 Materials

The materials utilized in corona discharge processes may lead to different results during the discharge, especially in terms of lifespan, maximum applied voltage, corrosion, etc. Each material has its own properties that determine its capability to perform a specific application.

In this project, the main materials used in corona discharge are, as it is said before, tungsten and aluminum. Because corrosion of the corona electrode is the main limitation of corona discharge, corrosion mechanisms of both electrodes have been already explained before. The device that is going to be subjected to corona discharge is represented in Figure 11 and it corresponds to a wire-to-plane configuration, where the corona electrode is the wire and the mass electrode is the plane.

Corona electrodes are made of tungsten filaments and there are two diameters used in the devices. Tested wires have a diameter of 100 μm (microns) or a diameter of 25 μm , in addition, both have 40 mm of length that is facing the mass electrode. According to the previous section, smaller diameters provides a higher current, which increases the corrosion, decreasing strongly the lifespan but gives more power. Mass electrodes are made of aluminum plates of 40x40mm, the plane geometry assure us that the ionization is going to be carried out at the corona electrode, which is more curved. Between both electrodes, there is a 2 mm gap where the air particles and species can move through. Furthermore, there are dielectric barriers at both sides of the corona electrode because of there are three tungsten wires per device, as observed in Figure 11, hence, dielectric barriers are necessary to avoid interaction between corona electrodes, so it does not affect the testing.

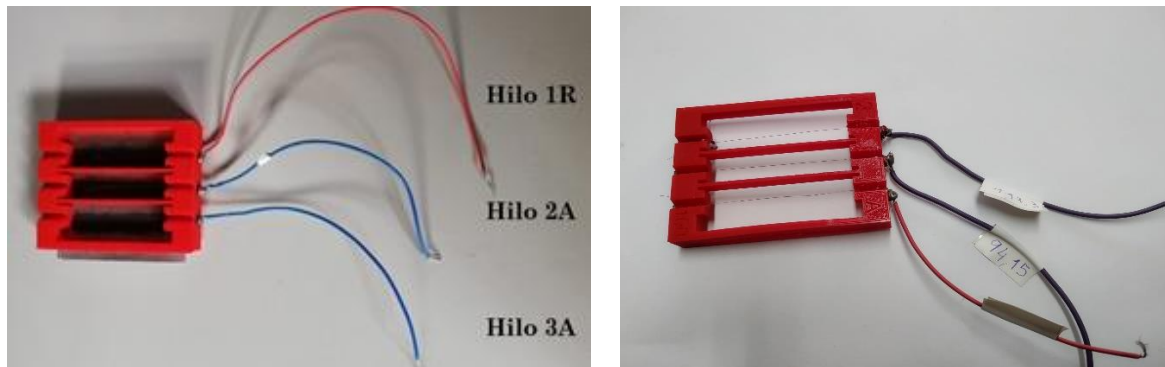


Figure 11. Corona discharge wire-to-plane devices. Both electrodes in the device (left)/ only corona electrode (right).

Once the device has been set up, it is subjected to corona discharge. A scheme of the process is represented in Figure 12, where the corona electrode is connected to the positive current through the wires welded to the electrode, whereas the mass electrode is connected to a smaller voltage or mass. After that, the current is applied during a variable time, going from 4 hours up to 120 hours of testing. Finally, when the test is done, the resulting corroded corona electrodes and mass electrodes are sent to our lab in order to study the corrosion undergone by the materials.

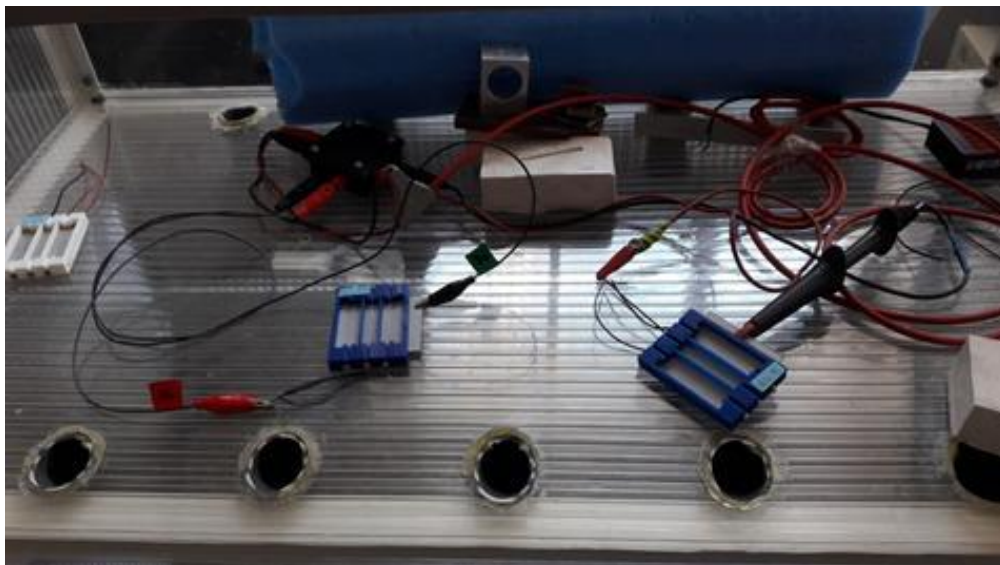


Figure 12. Corona discharge testing set-up.

4.2 Experimental and computational methods

In this section there is an explanation of the sample preparation to its subsequent analysis and the experimental methods utilized to observe and analyze the corrosion of the filaments and plates in order to, finally, discuss the results obtained and make conclusions.

4.2.1 Sample preparation

The first step in analyzing the electrodes is to prepare the samples for the following analysis methods. The sample preparation depends on the method that is going to be utilized subsequently. In this project, the sample preparation is focused on Scanning Electron Microscopy (SEM), which its function fundamentals are detailed in the next section.

Corona discharge devices are made of polymer, a non-conductive material, as observed in Figure 13, so in order to obtain a good image in SEM, it is necessary to place the filaments on a conductive material. Therefore, corona discharge devices need to be disassembled in order to get the filaments and place them on a prepared sample towards SEM. A screwdriver is utilized to remove the screws from the upper part of the structure presented in Figure 13 and scissors are used to cut the filament in the bottom part of the structure, obtaining the filament.

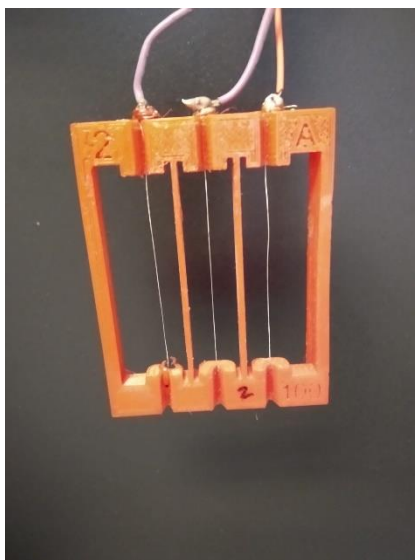


Figure 13. Corona discharge device with a polymer structure after testing.

The next point is to obtain the sample where the filament is going to be placed. With the help of design programs like SketchUp, a sample, represented in Figure 14, was designed for later mechanization in aluminum. The first design has a similar structure to corona discharge devices previously showed and has almost the same dimensions, on the other hand, the main difference is the lack of the dielectric barriers and the indent where the filament is placed, being shallower.

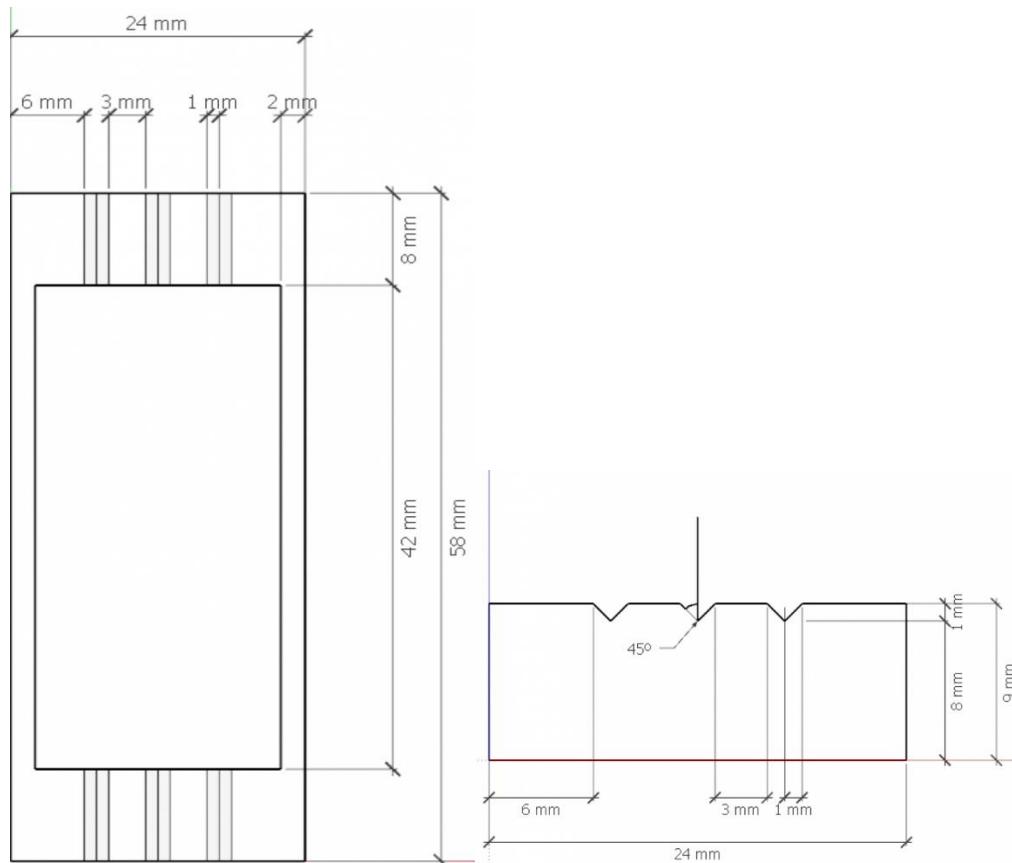


Figure 14. First design of the sample: top view (left)/front view (right).

After this first design, it was observed that the sample was bigger than necessary, furthermore, although the indent is only of 1 mm deep, the less deep the indent, the more efficient is the SEM when analyzing it. This is because the cabin where the sample is placed when is going to be analyzed within the SEM, is small and the filaments have to be as close as possible to the detectors.

Finally, after this discussion, it was concluded that the easiest way to analyze the filaments was to design a sample with no indents and C-shaped, as showed in Figure 15. Furthermore, this design is thinner than the first one and its structure is considerably simpler, making easier its mechanization.

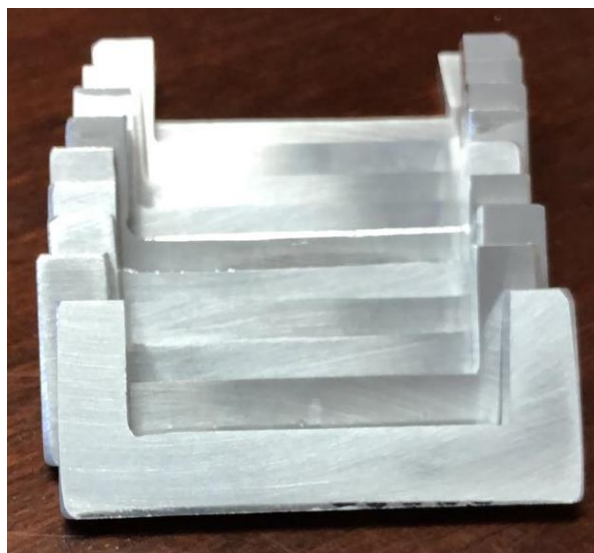


Figure 15. C-shaped samples for tungsten filaments.

Once the C-shaped samples have been mechanized, the next step is to clean the surface of the samples in order to increase the effectivity during SEM stage. Samples must be as clean as possible so the surface impurities do not affect the filament image. The cleaning of the samples is carried out by an ultrasonic cleaner.

Ultrasonic cleaning is a process where ultrasound is used to induce cavitation bubbles in a fluid agitating it ^[22], removing all particles of contamination from the surface of the desired object, the effect depends on the fluid. The C-shaped sample is placed in a glass with acetone, which is the fluid that is going to clean the surface of the sample. Following, the glass with the sample and the acetone is placed in the ultrasonic cleaner, where there must be water surrounding the glass to make sure that the vibration of the ultrasound reaches correctly the acetone and agitates it, cleaning the sample. The glass is subjected to ultrasonic cleaning for around two minutes. When the process has finished, the glass is retired and, subsequently, the sample is taken off the glass in order to, right after, dry the surface with a dryer and place them in a piece of paper. The process is repeated with all the samples.

After the cleaning process is done, it is necessary to add carbon tape to the sample. The carbon tape is a sticky and conductive material that is commonly used in SEM. Its sticky properties are really helpful in order to maintain the material fixed. Furthermore, the conductive properties of the carbon tape are necessary to avoid the charge accumulation on the material, otherwise, a charged-up material will look brighter and it will decrease the image quality. Finally, this type of sample, represented in Figure 16, is obtained and now, it is ready to be introduced in SEM in order to be analyzed.



Figure 16. C-shaped sample with carbon tape and tungsten filaments.

4.2.2 Scanning Electron Microscopy (SEM)

Scanning electron microscopy is an analysis technique invented by Max Knoll in Berlin in 1935^[23], but he did not bother to patent it. Some years later, Von Borries and Ruska at Siemens were the first to obtain a high-resolution image from Transmission Electron Microscopy (TEM) that, as a consequence, led to SEM become obsolete because it will never be able to compete with TEM image quality. This view changed when the field emission guns were available^[23]. Today, SEM is one of the most used techniques in many areas of science and technology. In order to understand its functioning, it is necessary to introduce the components of the microscope and the physics principles where it is based on.

SEM forms an image by scanning the surface with a focused beam of electrons working in vacuum. A scheme of the components of a SEM is represented in Figure 17 (a). The electron gun is the responsible to form the electron beam that is going to be used to analyze the sample. There are different materials used to form the electron beam, but the material used in this project is a tungsten filament in the SEM located in the laboratory of Escuela Técnica Superior de Ingeniería Aeronáutica y del Espacio (ETSIAE) from Universidad Politécnica de Madrid (UPM). The anode is used to attract the electron beam and the magnetic lenses are necessary to focus the electron beam as it moves down the column. In the chamber is where the sample is placed and the vacuum is formed prior to SEM functioning. Finally, the beam reaches the sample and it generates different types of electrons that are collected by the detectors.

Each type of electron generated during the process is formed from a different interaction between the electron beam and the sample and, therefore, provides different information of the surface when collected. There are primarily two types of electrons collected from the process: Secondary electrons (SE) and backscattered electrons (BSE).

The secondary electrons are formed by inelastic collisions between the beam electrons and the atomic electrons of the sample. Low energy beam electrons collide with the atomic electrons and, however the energy is low, it still have enough energy to release electrons, consequently, electrons can only escape if they are close to the surface, from 5 to 50 nm deep, and, therefore, be collected by the secondary electron detector. Because of the interaction is close to the surface, secondary electrons allow us to obtain information from the topography and the surface of the sample.

Backscattered electrons are formed by elastic collisions. Because of the energy of the incident electrons is higher, they emerge from deeper locations within the sample, between 50 and 300 nm in general. It provides less information about the surface than secondary electrons but BSE provide a distribution of elements in the sample and, used along with x-rays formed, are used to obtain information of the composition of the sample.

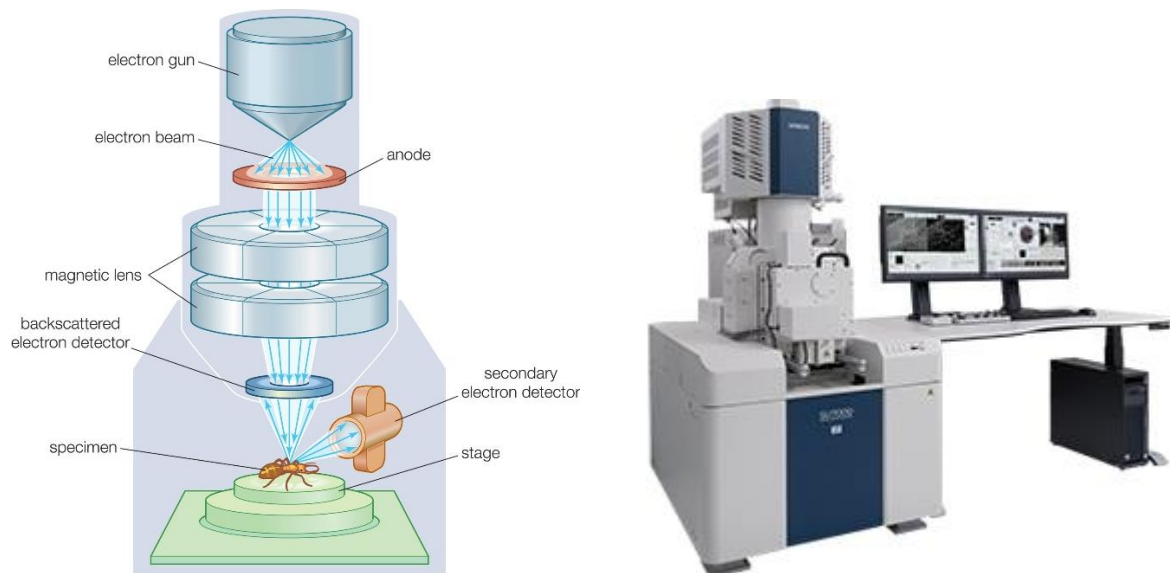


Figure 17^[22]. (a) Scheme of scanning electron microscopy components (left)/(b) Real scanning electron microscopy (right)

Finally, some materials considerations have to be taken into account. The higher atomic number elements contain a higher number of electrons in the atoms and, hence, emit much more than the lower ones, therefore, the higher atomic number elements are seen

brighter than the lower. Another consideration is the voltage used to illuminate the sample, that can variate in terms of kV, in this project usually a value of 15 kV is used. In addition, some conductive and non-conductive materials such as polymers does not emit electrons so easily, so first the material charges up and when it is charged enough, it releases a high number of electrons, leading to a bright image that interfere with the contrast of itself, decreasing its quality. That is why it is so important to use conductive materials and carbon tape in order to avoid the charge-up.

Once good images of the filaments are obtained, a discussion of them is going to be performed in order to evaluate de corrosion of the tungsten. Furthermore, with the aim of obtaining the composition of the surface and, therefore, obtain more information about the corrosion, the sample is going to be subjected to EDX.

4.2.3 Energy Dispersive X-Ray Spectroscopy (EDX)

Energy dispersive x-ray spectroscopy is an analysis technique used for chemical characterization of a sample. This type of analysis takes advantage of the x-rays emitted from the electrons of the sample. To obtain these x-rays it is necessary to induce energy in the atoms of the sample, by an x-ray beam or, in this case, the electron beam itself obtaining secondary and backscattered electrons as well. The resulting x-rays are collected by a specific x-ray detector.

In order to be more specific, x-rays are a phenomenon produced by the inner shell excitation with incident electrons. This means, when an inner shell electron moves to a higher energy state, the hole left behind the electron in the inner shell is filled by another electron at a high energy level, resulting in the emission of a characteristic x-ray between those two energy levels. The depth where this process can occur is up to 3 μm ^[24].

Because characteristic x-rays have specific energies corresponding to each element, the element can be identified from the peak energy, seen as an example in Figure 18. In addition, the content of the element in the compound can be analyzed from the integrated intensity of the peak. Energy peaks positions are predicted by the Moseley's law ^[24]. It has to be explained that when the excited state the atom with a hole changes to the ground state, an Auger electron may be emitted instead of the characteristic x-ray emission. In general, the x-ray emission probability increases with the increase in atomic number, conversely, the

Auger electron emission decreases^[23]. Thus, heavy elements will be easier to analyze than light elements by EDX process.

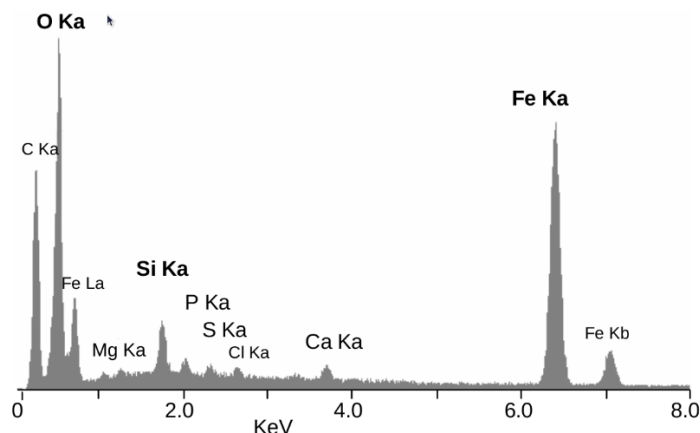


Figure 18. Example of energy peaks of a sample using EDX spectroscopy.

4.2.4 ImageJ^[26]

ImageJ is a free software focused on image analysis where it can display, edit, analyze and process images from 8-bit to 32-bit, furthermore, it can read many different image formats. It is multithreaded, so time-consuming operations such as image file reading can be performed in parallel with other operations.

This is a software that can perform different types of calculations on an image. It can calculate area and pixel value statistics of user-defined selections. It can measure distances and angles. It can create density histograms and line profile plots. It supports standard image processing functions such as contrast manipulation, sharpening, smoothing, edge detection and median filtering.

It does geometric transformations such as scaling, rotation and flips. Image can be zoomed up to 32:1 and down to 1:32. All analysis and processing functions are available at any magnification factor. The program supports any number of windows (images) simultaneously, limited only by available memory. Spatial calibration is available to provide real world dimensional measurements in units such as millimeters. Density or gray scale calibration is also available.

In this project, the use of ImageJ is focused to the calibration of the images obtained by SEM in order to measure some geometrical parameters such as the diameter of the filaments.

4.2.5 Computational simulation

In order to expand the perspective of the corrosion undergone by the materials during corona discharge processes, a simulation by the software COMSOL Multiphysics is performed.

COMSOL is a software widely used when simulations of physical phenomena are required, such as fluids, electricity, plasma, corrosion, etc. Because of its complexity and equations, COMSOL is a program with an outstanding accuracy level of simulation. As it is said in the previous chapter, the equations and the simulation carried out by this program are focused in corrosion, taking into account the current and mass loss conditions produced by corona discharge.

Now, the simulation modelling is going to be shown with the aim of understand the functioning of the program. The procedure is divided in different steps, always following the same sequence. Although there are many steps, they are going to be grouped up into three sections.

4.2.5.1 Geometrical model

The first step in COMSOL Multiphysics is to create the geometry of the sample that is going to be corroded. This geometry must reproduce as good as possible the original device subjected to corona discharge. In the upper part of the interface of the program, click on add component, and a 2D model is selected. A 3D model is not performed because of the high computational cost of that kind of simulations compared to the available computers in this project.

Following, the 2D geometry has to be created. First of all, the corona electrode geometry is defined, providing the radius of the filament, 50 μm in the case showed in this section, the x and y coordinates, where the y position is set to 1 mm. The next step is to define the geometry of the mass electrode. That is represented with a height of 0.15 mm and

a width of 4 mm, and located at y position equal to -1 mm in order to create a gap of 2 mm between the electrodes. Finally, the electrolyte has to be defined. In order to do so, a rectangle of 5 mm of width and 3.5 mm of height is created at the position $y = 0.675$ mm as shown in Figure 19.

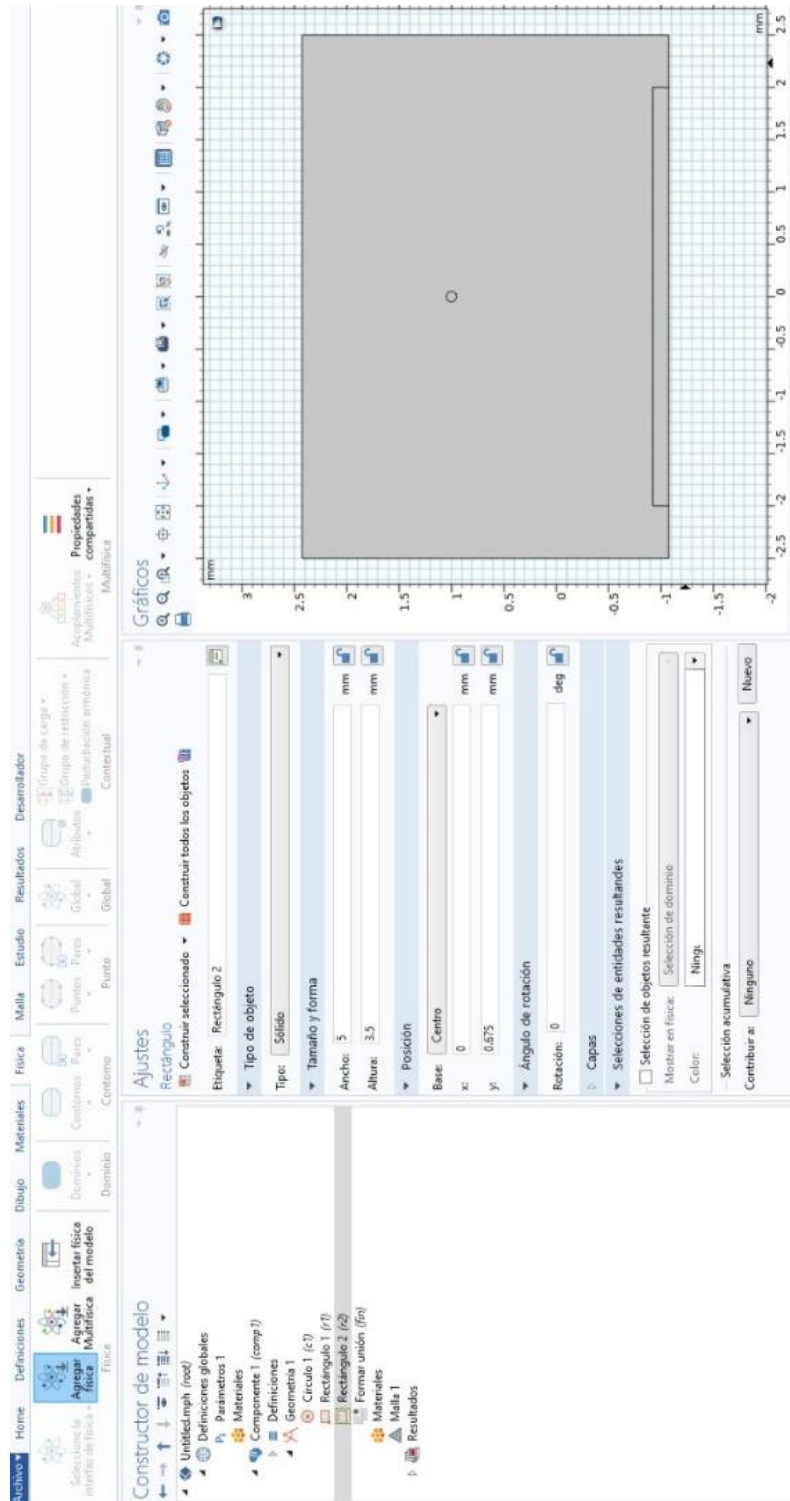


Figure 19. COMSOL geometry creation interface.

The final part of this subsection is to define the materials of each geometry previously created. On the superior part the add material button is clicked and, following, the materials are searched and selected from the database of the program. The materials are: Air for the electrolyte, represented with the biggest rectangle, tungsten for the corona electrode, represented with the circle and, finally, aluminum for the mass electrode, represented with the small rectangle.

Once the geometry and materials of the simulation are defined, it is time to add the corresponding physics.

4.2.5.2 Physics

Click on the physics section, add physic. On the database the physic of interest is searched, in this case, electrochemistry, corrosion and deformed geometry. Once here, secondary distribution is selected. Tertiary is not selected because it is more focused on liquid electrolytes and galvanic cells, furthermore, this corrosion module requires more computational cost and time. Once the physic is selected, notice that deformed geometry and multiphysics are also added. Deformed geometry is focused on the deformation undergone by the electrodes when corroded, and the multiphysic correlates corrosion and deformation physics.

First of all, parameters of the electrolyte have to be defined. As shown in Figure 20, the temperature is user defined, applying room temperature (293 K). Below the conductivity of the electrolyte has to be defined. Because the electrolyte is air, the corresponding conductivity is $5\text{e-}15$ S/m.

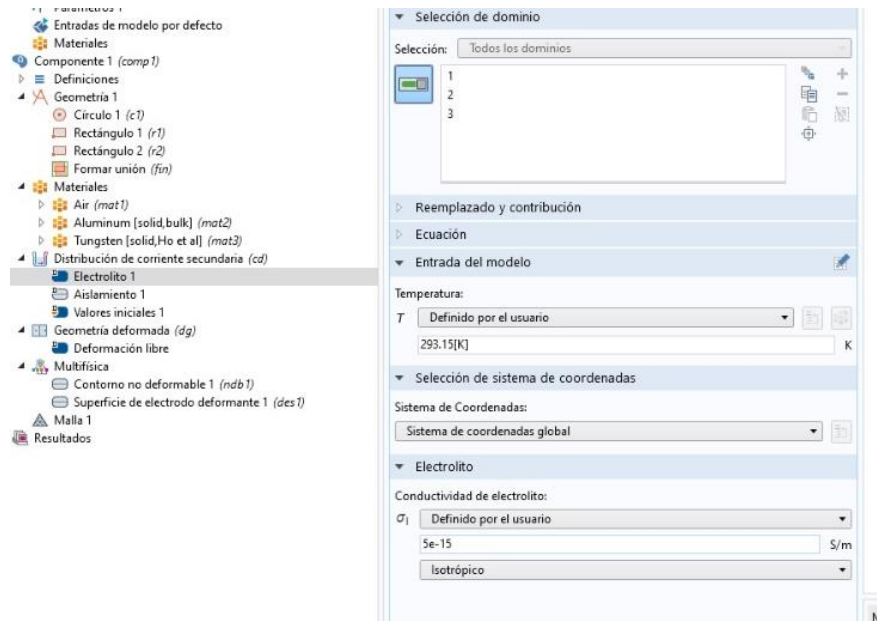


Figure 20. COMSOL definition of electrolyte parameters interface.

Once electrolyte parameters are defined, boundary conditions shall be applied in order to determine the parameters of the electrodes. First of all, click on boundary and the electrode surface boundary condition has to be added. Once this condition is added, the corona electrode surface is selected for this condition. The interface is represented in Figure 21, where some parameters must be defined. In dissolved species, tungsten has to be added as well as its density and molecular mass, both parameters shown in Figure 21. In the boundary condition in the bottom part of the interface, electrical potential shall be applied to simulate the electric current moving through the electrode, this value is set to 5 kV if the diameter of the filament is 100 μm and 4 kV if the diameter is 25 μm .

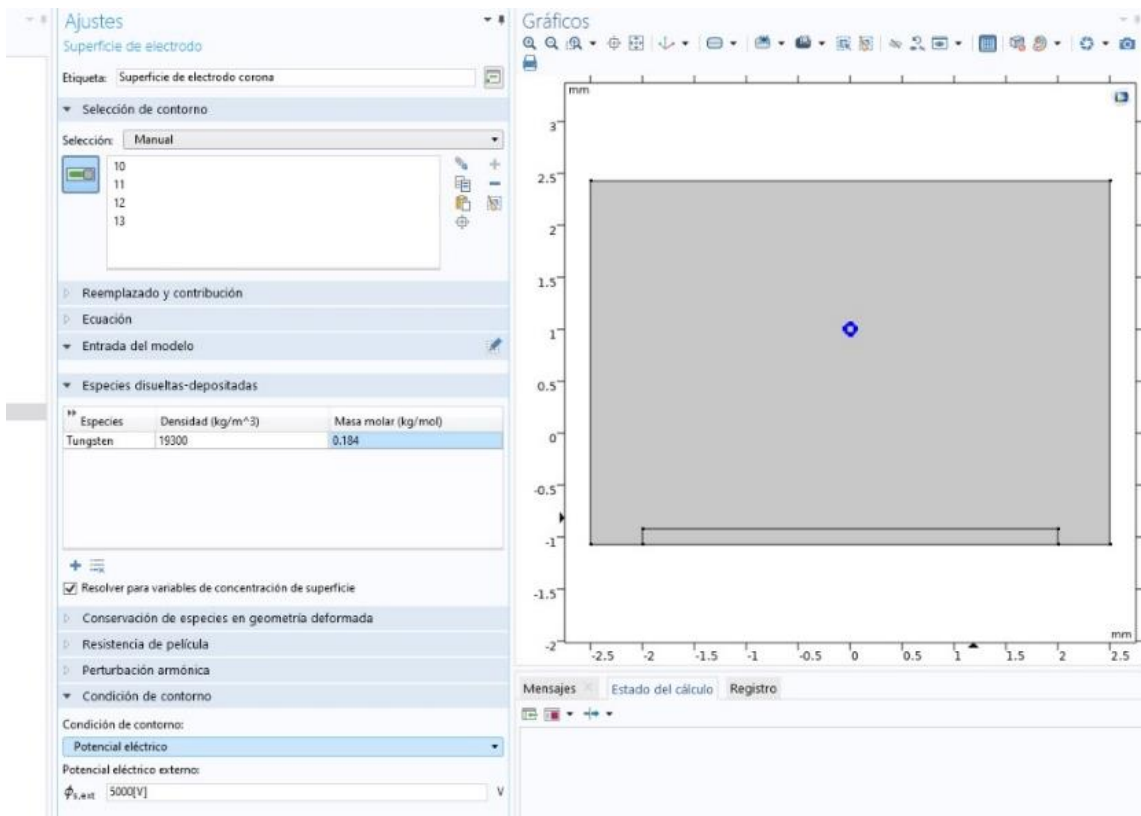


Figure 21. COMSOL electrode surface boundary condition interface.

Following, within the electrode surface boundary condition, further parameters can be found, shown in Figure 22. This subpart is called electrode reaction and is used to define basic reaction parameters during the corrosion of the electrode. The temperature is user defined to room temperature. The number of electrons involving the formation of tungsten trioxide is 6 electrons and the stoichiometric coefficients is 1. Below, the equilibrium potential is user defined and the value is equal to -0.58 V, that correspond to the potential of tungsten oxidation to form WO_3 . Finally, the kinetic expression is set to Butler-Volmer as an approximation of atmospheric corrosion, neglecting anode and cathode reactions.

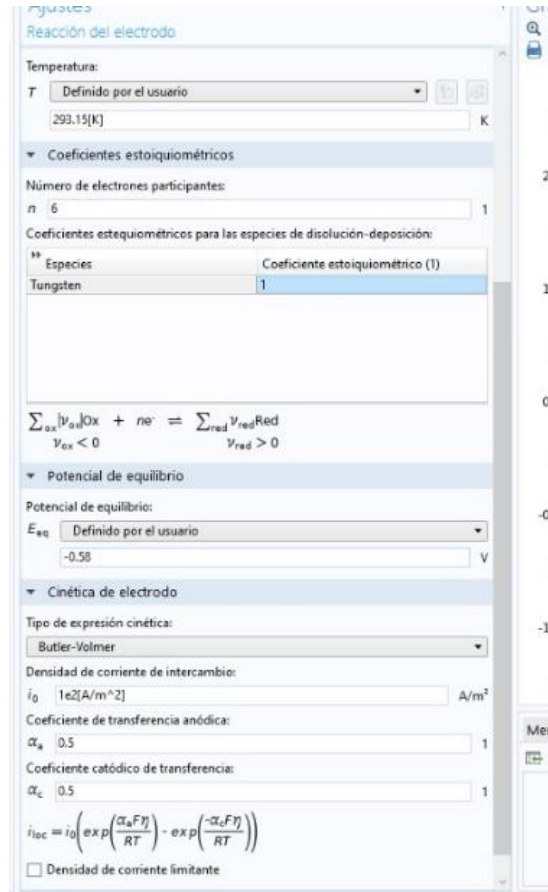


Figure 22. COMSOL electrode reactions interface.

The next step is to define the electrode surface of the mass electrode. Following the same procedure as the corona electrode and particularizing the parameters to the mass electrode. In dissolving species aluminum is added with a density of 2700 kg/m^3 and a molar mass of 0.027 kg/mol . The electrical potential boundary condition is selected and set to 0 V .

Following, the reaction parameters of the mass electrode are defined. With the number of electrons involved in the reaction equal to 3, the stoichiometric coefficient equal to 1 and the equilibrium potential equal to -1.66 V , these parameters define the alumina generation from aluminum oxidation. Finally, the kinetic expression is also set to Butler-Volmer.

Once the boundary conditions of both electrodes are defined, further boundary conditions regarding electrolyte must be defined in order to make a more accurate simulation. As shown in Figure 23, the infinite electrolyte boundary condition is defined on the edges of the larger rectangle, so the program neglect the larger rectangle as a limit of the electrolyte. In addition, the conductivity of air has to be defined again in this boundary condition, being the value equal to $5\text{e-}15 \text{ S/m}$.

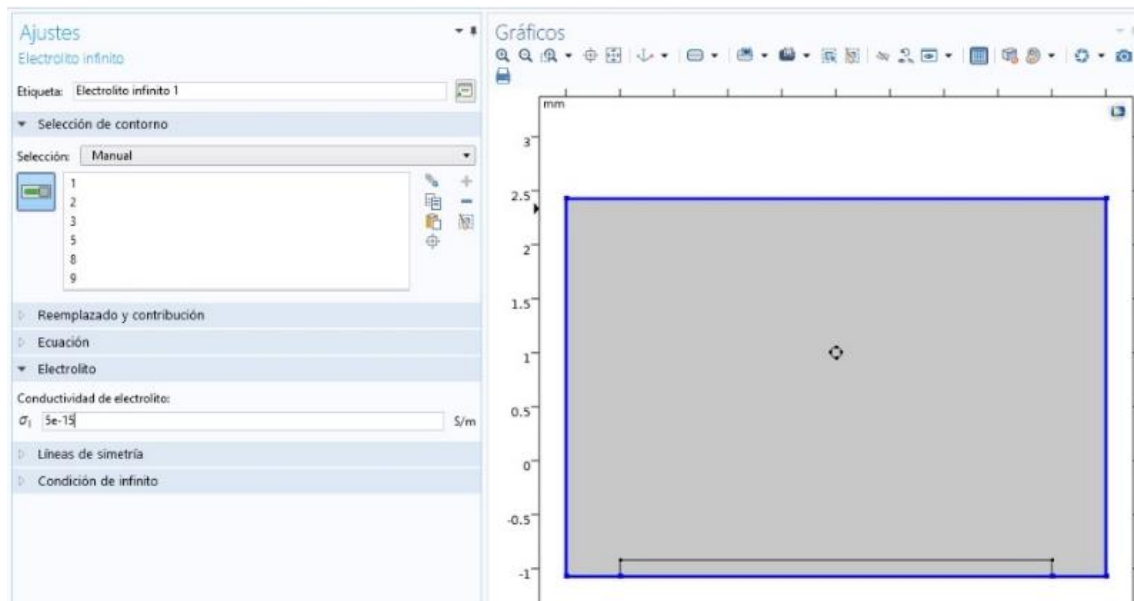


Figure 23. COMSOL infinite electrolyte boundary condition interface.

Finally, the last boundary condition is called electrolyte current density, that will be the parameter used to simulate the corrosion undergone by the electrodes. The thickness change along the electrode surface depends on the current density through electrolyte, so this value is very important for an accurate simulation.

In order to introduce the parameter of current density through electrolyte in the model, the surface of corona electrode is selected. Following, a value of current density has to be added. This value is the intensity that leads to corrosion from the equation presented below:

$$w = \frac{ItM}{nF} \quad (31)$$

Being w the mass loss, I the current, M is the molecular mass of the material, n is the number of electrons, t is time and F is the Faraday constant. Experimentally, the mass oxide per unit of area has been calculated both for $100 \mu\text{m}$ and $25 \mu\text{m}$. Applying the value of mass oxide per unit of area the current density can be directly calculated.

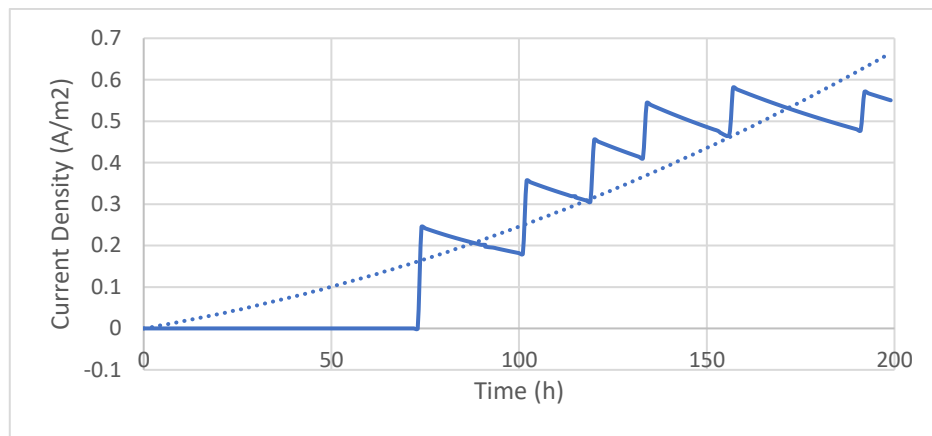
The value of mass oxide per unit area of a $100 \mu\text{m}$ corona electrode is determined experimentally by CEDRION, that subject the filaments to corona discharge. The results obtained from all the same type filaments are collected, subsequently, the average of the results is exposed in excel. Using those values and applying Equation 31 with the values of Table 1, the values of current density are obtained for every time step in hours.

Molar mass (kg/mol)	0.184
Number of electrons	6
F (Faraday constant (C/mol))	96500

Table 1: Tungsten data for Equation 31.

Once the values of current density are obtained in excel, the graph of the values is plotted, as shown in Figure 24. The tendency line is also added in order to make a numerical approximation of the obtained graph. The values of the graph are up to 200 hours, which is longer than any filament tested in this project. The tendency line is approximated as a second-grade polynomial expression. This expression is represented below:

$$I = 9e^{-6}t^2 + 0.0016t \quad (32)$$

**Figure 24.** Tungsten 100 μm current density vs time graph with polynomial tendency line.

Therefore, Equation 32 is going to be used as an approximation in order to calculate the current density that depends on time. This equation is going to be inserted in COMSOL and added as the value of current density through electrolyte on the corona electrode surface.

The first step to add this equation is to define its parameters with the correct units, otherwise the program will generate an error on the calculation. So, as shown in Figure 25, the added parameters correspond to the polynomial coefficients where A is the t^2 coefficient and B is the t coefficient. It is important to remark the units of the parameters, that are set to $1/\text{s}^2$ and $1/\text{s}$ respectively. Moreover, an additional parameter called ‘icor’ is introduced. Its only function is to fit the units of current density. The value is set to -1 because the current density must be negative to generate corrosion in the program.

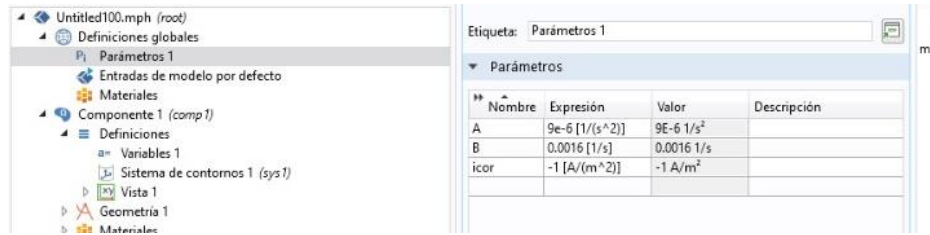


Figure 25. COMSOL global parameters interface.

After the parameters have been inserted, now it is time to define the equation in the program. As shown in Figure 26, two variables are inserted in local definitions. The first one corresponds to the current density of corona electrode and the second one to the mass electrode. The corona electrode equation is inserted with the particularity that the variable 't' is divided by 3600 because the program has 't' predefined to work in seconds, whereas the parameters of the equation are set to work in hours.

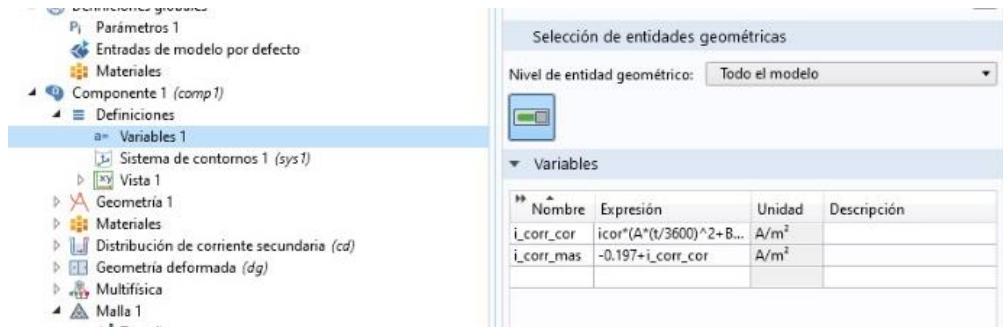


Figure 26. COMSOL local variables interface.

In order to make an approximation of the mass electrode current density, the fact is that this electrode undergoes more corrosion due to the air and ozone flux that flows towards it. So, the value of ozone corrosion of aluminum has been calculated using Equation 31 and a value of 0.066g/m²h^[27]. The current density result is equal to 0.197 A/m², value that is added to the corona electrode current density in order to take into account the corrosion induced by corona discharge and the corrosion induced by ozone and air flow. Therefore, the current density is equal to the second equation observed in Figure 26.

The same process is performed for 25 μm corona electrodes, obtaining Equation 33.

$$I = -2e^{-5}t^2 + 0.0026t \quad (33)$$

Now that the equations are obtained and inserted in the program, it is time to set the current density through electrolyte values equal to the equations. As represented in Figure

27, insert the equation is pretty simple, just put in the text box the equation label, in this case 'i_corr_cor'. The program will compute the calculations using the given equation varying it through time.

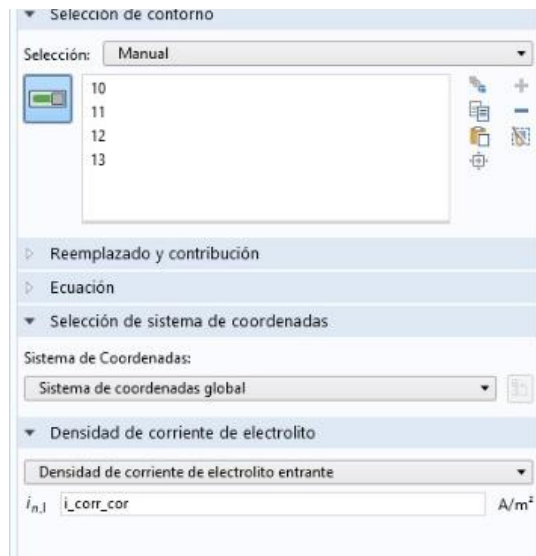


Figure 27. COMSOL electrolyte current density boundary condition interface.

The mass electrode current density in 25 μm corona electrodes is approximated in the same way as 100 μm . The previously calculated value of -0.197 is added to Equation 33 and inserted in the model, obtaining 'i_corr_mas'. Now, the secondary current distribution module has been completely defined.

The deformed geometry and the multiphysics are not modified because the software automatically fit deformed geometry physics to corrosion module, but it can be modified if deformation boundary conditions have to be applied.

Finally, in order to calculate the mass loss induced by erosion, the expression is going to be added to COMSOL so it can simulate it at the same time as it simulates the corrosion of the electrodes.

4.2.5.3 Mesh and study

Once the simulation is defined, the next step is to create a mesh that defines the accuracy of the program simulation with respect to the real life. Making a good mesh is very important because it increases the accuracy of the simulation optimizing the computational costs. The most important surfaces here are the electrode surfaces, therefore, the mesh has

to be focused on obtaining the best accuracy on these surfaces, while the rest of the electrolyte is less important. It is concluded that the mesh has to be thinner when close to the surface of the electrodes and thicker in the rest of the region.

The mesh has been created by adding three different edge mesh, one for corona electrode and the other two for mass electrode. In the corona electrode edge mesh, a distribution is added with 95 elements on the surface. In the two edge meshes for mass electrode, one of them is on the top of the geometry, facing the corona electrode. This first mesh has a distribution with 500 elements. The second mesh is related to the left and right part of the mass electrode, with a distribution of 30 elements. Finally, a free triangular meshing is added and the size selected is thinner, leading to the mesh showed in Figure 28.

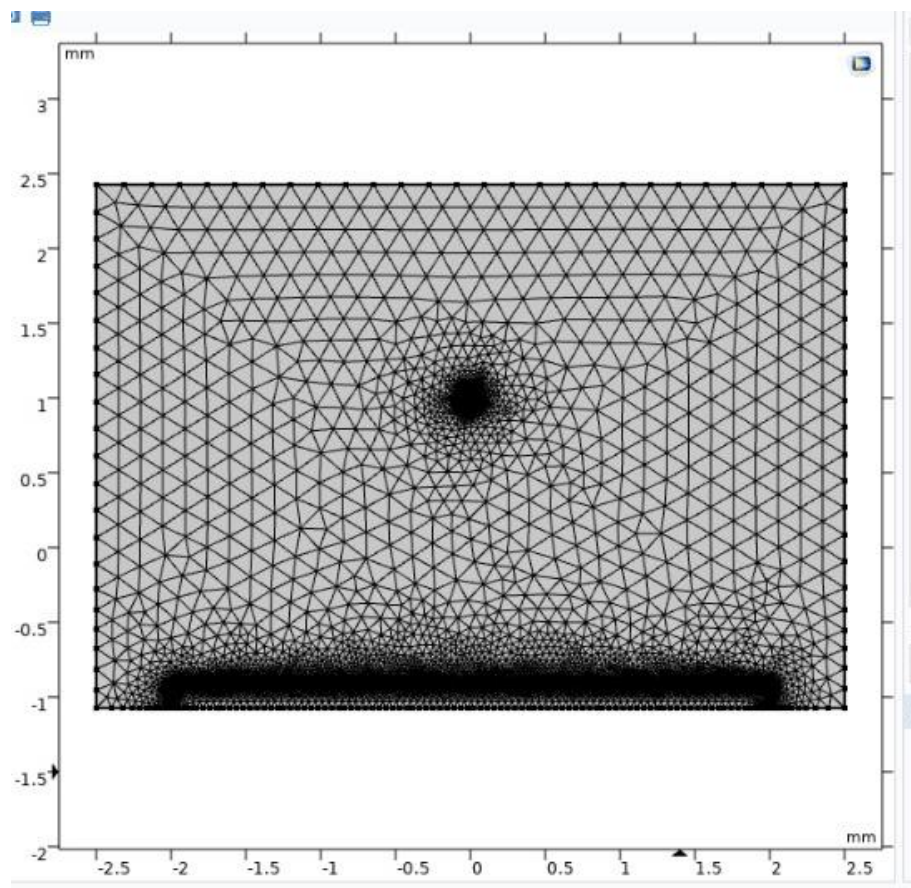


Figure 28. COMSOL model meshing.

The resulting mesh is composed of 30132 domain elements, 1136 boundary elements and, finally, 49798 degrees of freedom.

The study is also an important feature of the simulation, because the results obtained depend on what type of study is performed in the program with the existing model. Many studies can be performed, some stationary or time dependent. To select the study for this

simulation, click on study and add a time dependent with initialization. Notice that the study is recommended by COMSOL for our physics. As shown in Figure 29, some parameters must be adjusted. In the time dependent part of the study, a time of 180 hours is applied with a step of 1 hour, so the study goes from 0 to 180 hours in steps of 1 hour. Furthermore, click on the results while solving option so the program provides a graph of all the results in every step of time done.



Figure 29. COMSOL time dependent study interface.

Finally, many graphs can be obtained once the study has been performed and the program has done all the calculations. These graphs can be exported and represented in excel. This is shown in the following chapter, results and discussion.

5 Results and discussion

In this section, the results of the experimental methods and the computational simulation are exposed. Furthermore, after the results are exposed, a discussion of the experimental and computational results is performed as well as a comparison between them in order to obtain considerable conclusions of the corrosion.

5.1 Experimental results

The experimental results are those related to the experimental method, in other words, the results obtained by SEM and EDX. Results obtained by SEM are images of the corona electrodes, otherwise, the results of EDX are graphs of chemical composition of the selected area or point of the filament.

The way the filaments are analyzed is very simple. At least 3 images of each filament are obtained; one image of the top part, the second in the center and the third in the bottom part of the filament. In addition, eventually two more images of the filaments could be obtained depending on how much information can be obtained from those parts.

First of all, in order to have a general vision of the samples and its characteristics, a chart containing a summary of the characteristics of corroded tungsten filaments is introduced below.

As it is shown in Table 2, all filaments of 100 μm undergo a 5kV voltage whereas the 25 μm filaments undergo 4kV. The filament 1.5 is broken, which is very interesting to find out why. Conversely, filaments 1.1 and 2.1 are new, and are going to be used as a reference to compare the corrosion undergone by the filaments when subjected to corona discharge with different test times.

Filament	Diameter (μm)	Voltage (kV)	Time (h)	State
1.1	25	0	0	New
1.2	25	4	4	Normal
1.3	25	4	8	Normal
1.4	25	4	50	Normal
1.5	25	4	90	Broken
2.1	100	0	0	New
2.2	100	5	4	Normal
2.3	100	5	8	Normal
2.4	100	5	40	Normal
2.5	100	5	94	Normal
2.6	100	5	94	Normal
2.7	100	5	117	Normal
2.8	100	5	182	Normal
2.9	100	5	182	Normal

Table 2. Characterization of corroded tungsten filaments.

5.1.1 Filament 1.5

This device is very interesting because the filament was working for 90 hours until breaking and it can provide information about the corrosion mechanism responsible of this failure. This device only has one filament of 25 μm , therefore, two images of the filament are going to be exposed, one referring to the transversal view of the failure and the other one regarding the longitudinal view.

5.1.1.1 Transversal image

The transversal image, represented in Figure 30, was quite difficult to obtain because of two main problems. The first one is that, as it is observed in Figure 30, the image is not totally transversal. Placing the broken filament in a suitable position to obtain the transversal image was extremely hard, due to the low thickness, which make the filament hard to handle, furthermore, the tungsten filaments are rigid and bending the filament with the aim of finding a suitable position without breaking it becomes a really hard task. The second problem found when obtaining this image is the drift effect of SEM. When a small sample, in this case filament, is introduced in SEM and the electron beam is focused on a very particular area, for instance, when increasing the microscope magnification at x1000 or x2000, the focused

area charges up of electrons and, eventually, starts swinging making very difficult to obtain a clear image of the region.

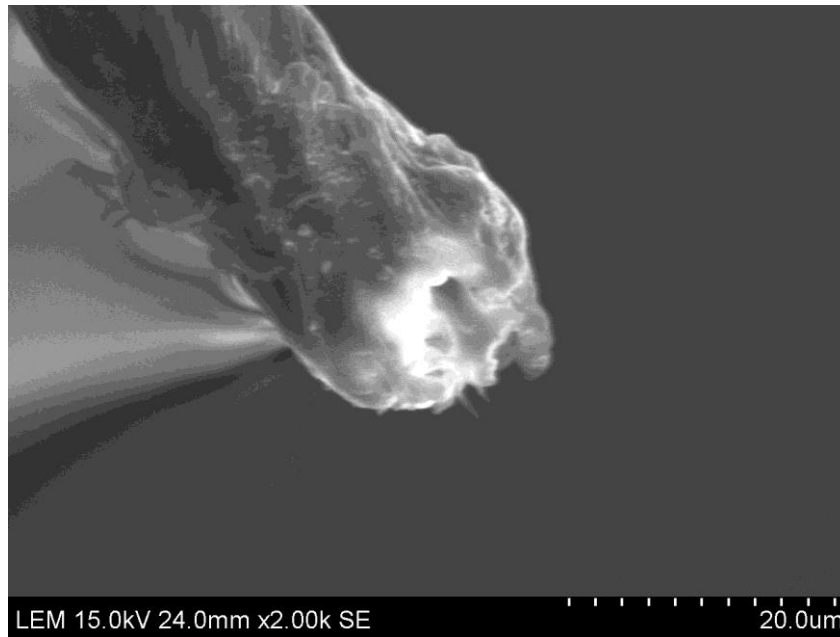


Figure 30. SEM transversal view of a broken 25 μm tungsten image, x2000 and 15kV.

In this image it is observed the corrosion of the surface of the filament as well as the corrosion of the inside in the cross-section cut. Unfortunately, this is one of the clearest images of the transversal view of the failure that was obtained. In order to observe a better view of the failure region, the longitudinal view is exposed following.

5.1.1.2 Longitudinal image

The longitudinal view, represented in Figure 31, is a very clear image where some deductions can be made. As it is observed, the image is very different to the transversal view, providing much more information. First of all, a neck is observed that indicates that the failure of the filament has not been fully brittle as it expected. Tungsten has a DBTT between 200 and 400°C, depending on metallurgical conditions. Therefore, its behavior should have been brittle., In other words, it is a partially ductile failure, because of necking observed under microscope reveals a ductile fracture character. The orientation of the observed lines in the top image leads to think that they are due to tension applied in the filament during set up. The filaments should be mounted as straight as possible prior to the corona discharge test, which requires to apply tension. Creep failure is not considered because of high melting temperature of tungsten and that the process takes place at room temperature and, at this

range, plastic deformation is only carried out by dislocation glide into the crystal structure, in this case BCC. However, plastic deformation requires the material works above its DBTT, therefore there could be a local increase of temperature.

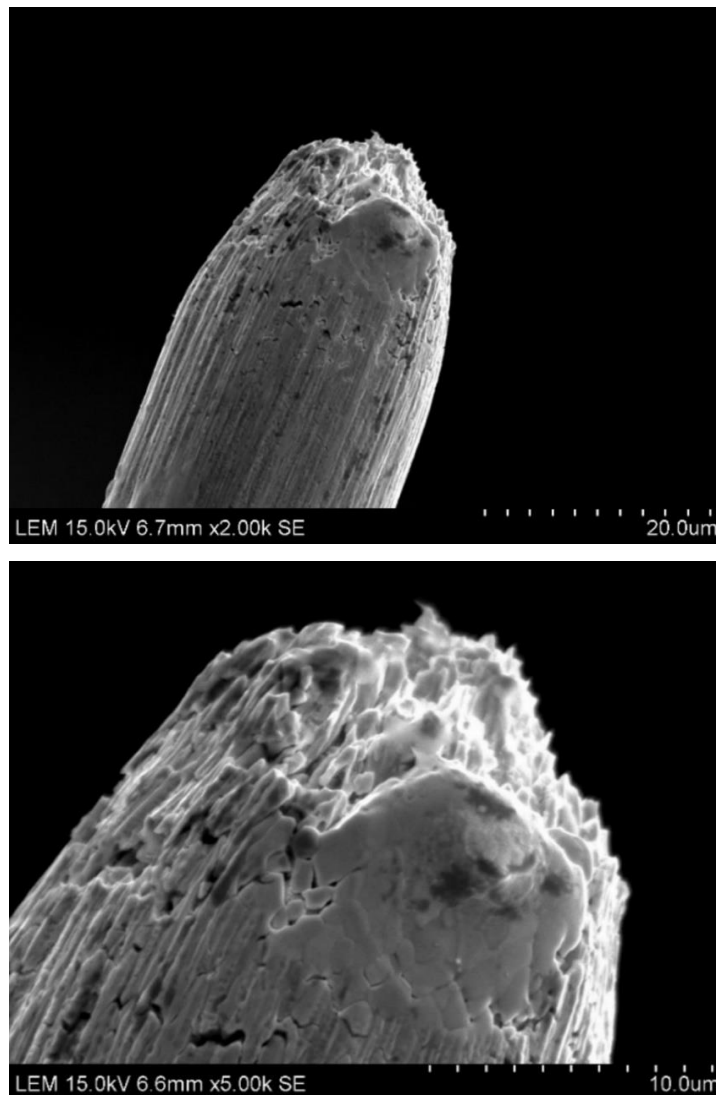


Figure 31. SEM long. view of broken filament of device five, x2000 (top image)/x5000 (bottom image).

There is no considerable corrosion as compared to other devices, but it is observed a possible intergranular failure, especially in the bottom image, where the decohesion of grain boundary can be seen. Intergranular fracture can be produced by a combination of stress, tension applied to the filament during the device set-up, and corrosion. This mechanism of failure is called SCC failure, but it shows a brittle character, non-coherent with necking in the fracture. But a combination of several mechanism can be considered.

Furthermore, using the ImageJ software in order to perform additional analysis on the image, a measure of the filament diameter appearing in the top image has been done by

calibrating the image with the scale in the lower right part of the image and, following, setting a segment that covers the whole diameter in a transversal direction. Finally, the program makes the calculation, resulting a value of $24.2\text{ }\mu\text{m}$, concluding that seems to be volume reduction, probably because of the mass loss produced by the corrosion and the erosion of the electrode. To conclude, the failure may be produced by the stress applied to the filament assisted by atmospheric corrosion and mass loss, both increased by corona discharge that, in addition, may have produced local heating.

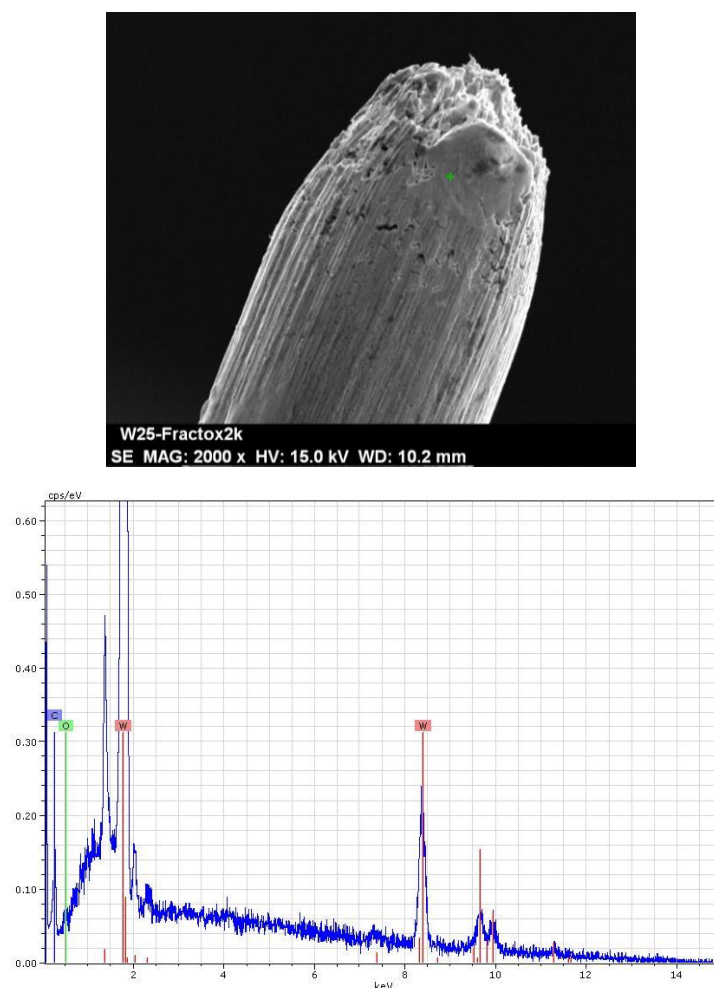


Figure 32. EDX 1 of broken tungsten filament of device five x2000, spot (top image)/EDX (bottom image).

Next to the fracture surface a circle shaped mass can be distinguished, that can be involved in the fracture. Microanalysis of this sample was performed by EDX in order to determine the composition of certain spots of the filament with the aim of find out the mechanism responsible of the failure. Composition can provide worth information related to corrosion due to the accumulation of elements. But it must be considered that the results are not very accuracy because of the small dimension of material. The first spot is represented in Figure 32.

As it is observed in EDX 1, the tungsten peaks are high because is the material of the filament, in addition, although the oxygen and carbon are not measurable elements by EDX, they are detected and must be considered when the peaks are very high or when compared with the same image but different spot. The second EDX of the three performed is represented in Figure 33.

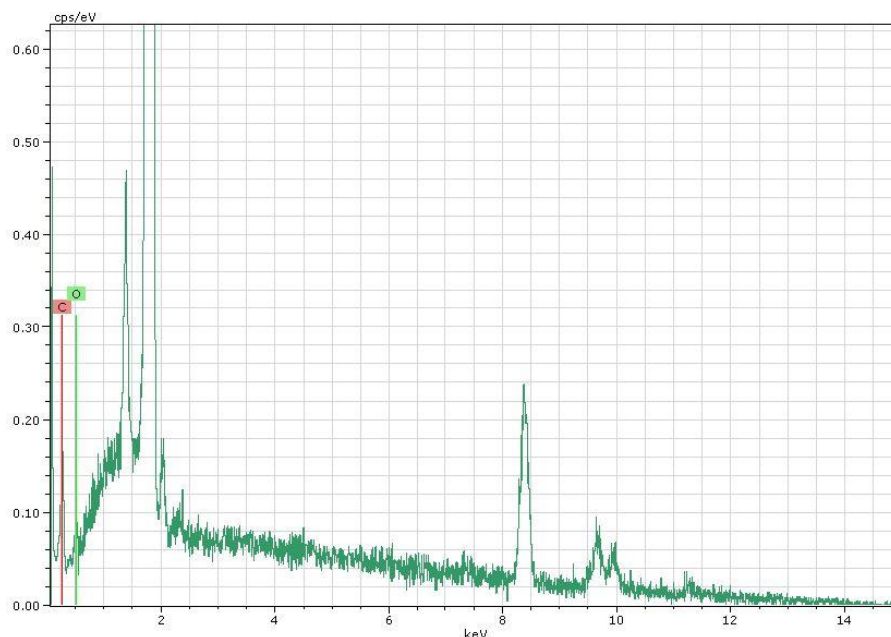
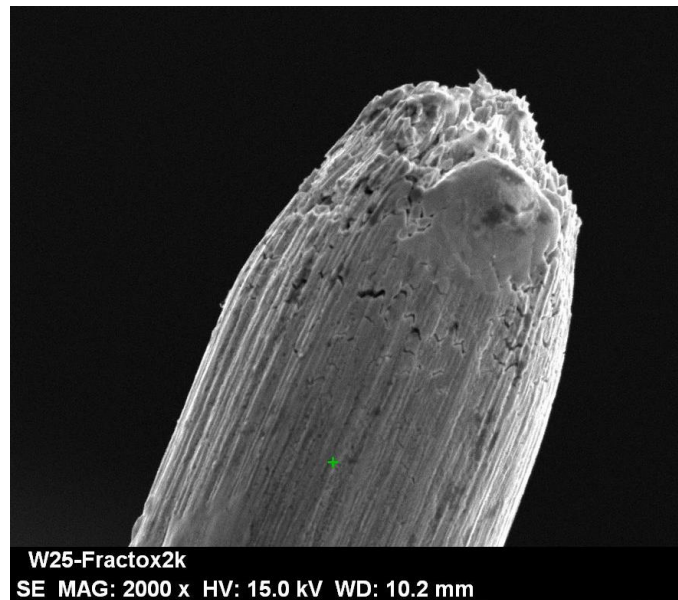


Figure 33. EDX 2 of broken tungsten filament of device five x2000, spot (top image)/EDX (bottom image).

This EDX has the same peaks when talking about tungsten but the carbon peak is lower, whereas the oxygen is almost the same. As it is said before, carbon and oxygen similarities do not provide any conclusion, only big differences can provide relative

information between two regions. The third EDX is more interesting and it is represented below in Figure 34.

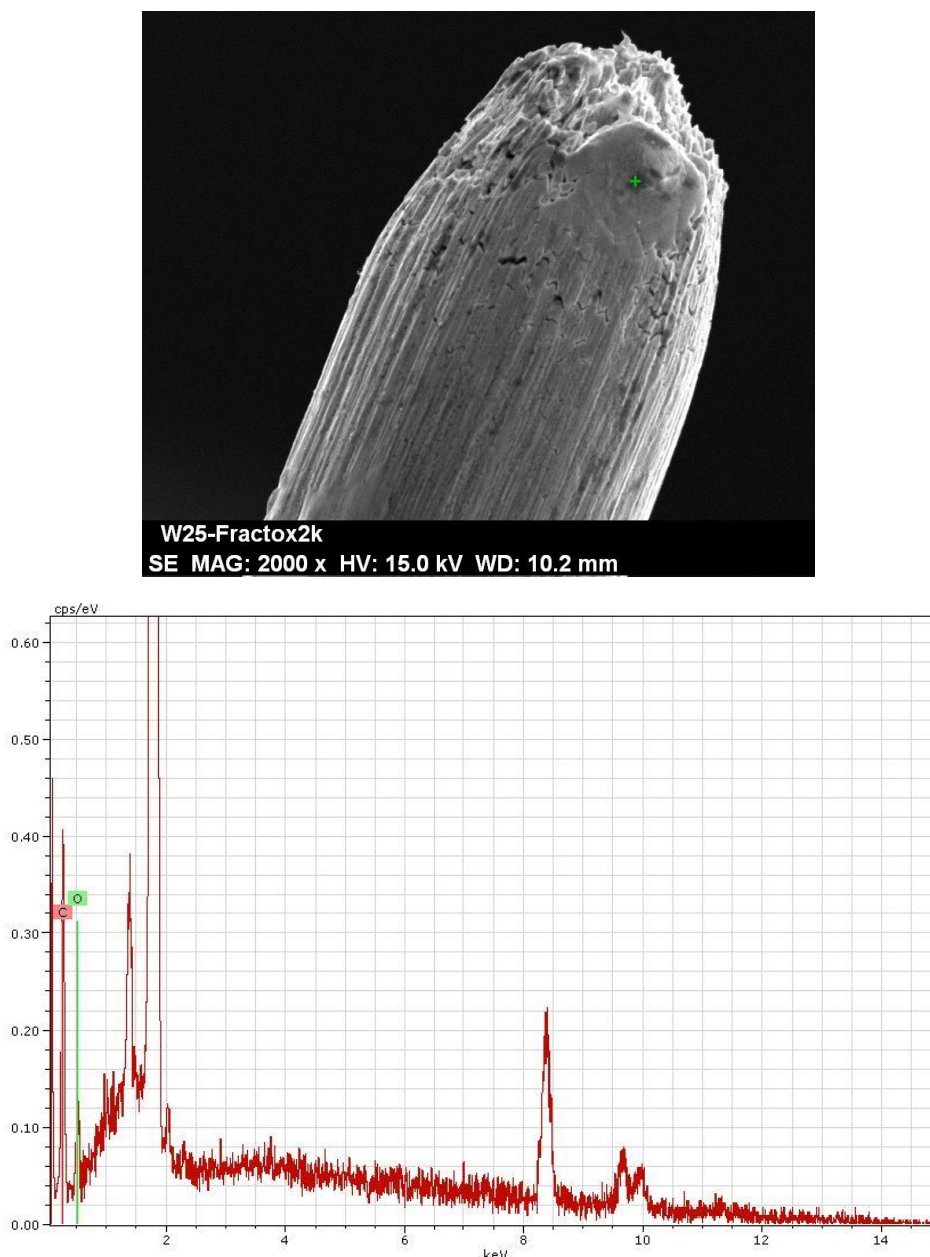


Figure 34. EDX 3 of broken tungsten filament of device five x2000, spot (top image)/EDX (bottom image).

The interesting point in this image is that the EDX has been performed on a dark region of the filament, leading to some differences in composition compared to the other two. As we can observe in Figure 33, the tungsten peaks are similar to the others EDX, but, conversely, the carbon and oxygen peaks are considerably higher, especially the carbon peak. In conclusion, the dark regions correspond to carbon or carbon compounds accumulations, as seen in chapter 2, this compound may be tungsten carbide. Finally, this leads to assume that carbon is not the only element that may accumulate in the surface of the

material, so accumulation of other compounds on the surface must be considered in order to enhance the efficiency of corona discharge.

5.1.2 Filament 2.4

The filament 2.4 has a diameter of 100 μm and a time of 40 hours, and is going to be analyzed in three parts as well as all the filaments, but, in this case, the parts to analyze are center, inferior and superior.

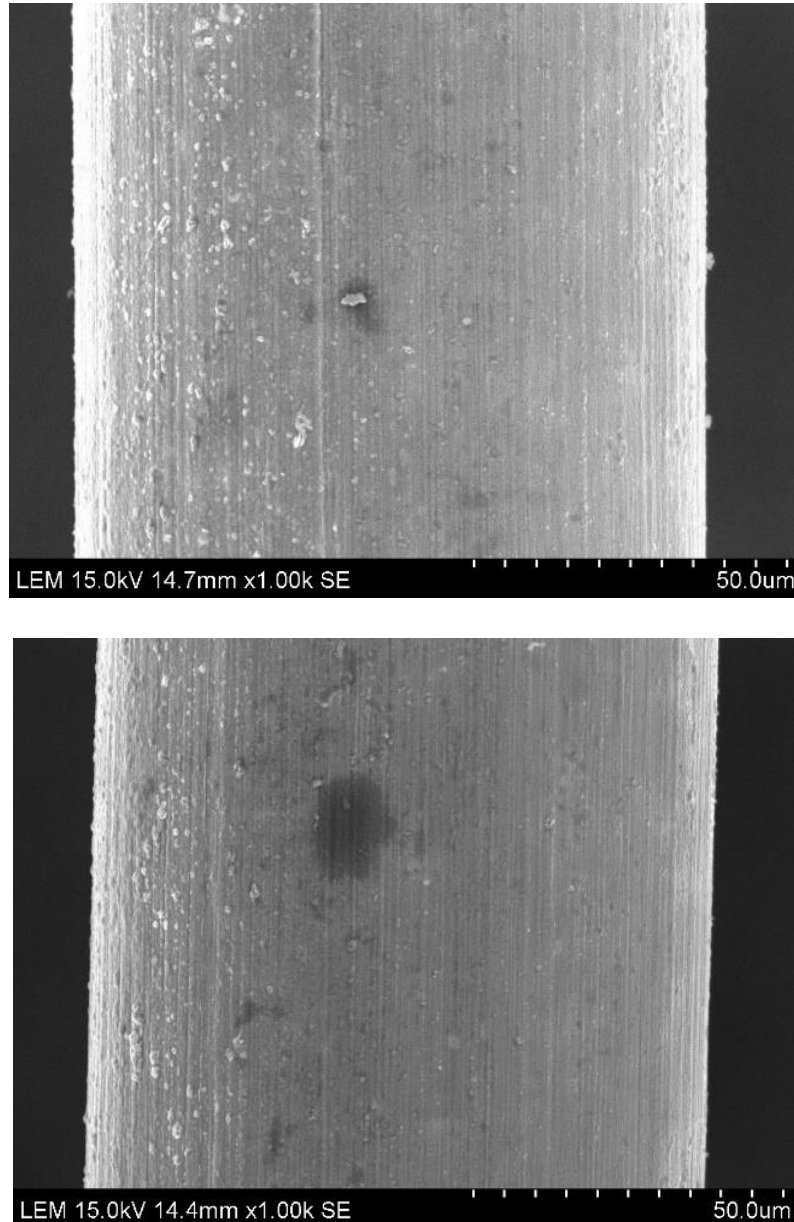


Figure 35. SEM images of center part of filament three from device four, x1000

The center part of the filament is shown in Figure 35, there are two images considered to contain information from the center part. It is observed that there are no big clues of

corrosion, however, the corrosion observed is located in the left part of the filament, because, as it is supposed before, that part was facing the mass electrode during the corona discharge process. Furthermore, it is also observed some black stains that may be produced by carbon accumulation or some oxides, and the lines along the Y-axis indicate the direction of the stress applied when the devices were set up.

The next part to be introduced is the inferior part of the filament, which is represented in Figure 36, where it is observed a different micrograph. What is observed right in the middle of the image are probably impurities, because of its shape. The black stains may be probably due to impurities presence or an accumulation of carbon.



Figure 36. SEM image of inferior part of filament two from device four, x1000.

Finally, the last image of this filament corresponds to the superior part and it is represented in Figure 37, where the oxide formation can be observed more clearly than the other two parts of the filament. Furthermore, this part seems like exposed to erosion because of the surface aspect shown in the image

In conclusion, in this filament it is observed a little more corrosion in the center part than the superior and inferior parts of the filament. Furthermore, all the images have been subjected to image analysis with ImageJ and the result has been quite surprising. The less corroded parts have reported a diameter of 98 μm (inferior) and 99 μm (superior), which differs from the reported in the center part, that is close to 100 μm . But these small differences can be considered as logical variation of filament dimensions due to manufacturing process.

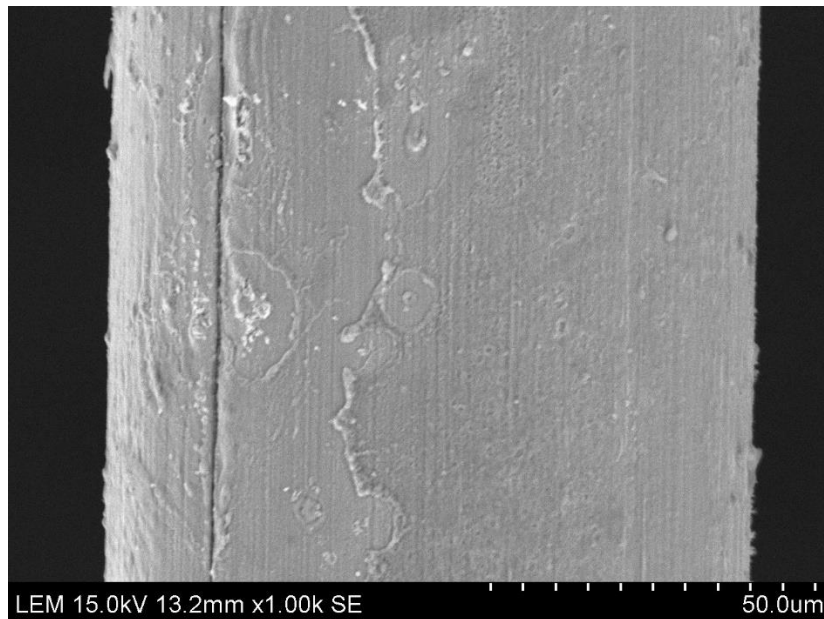


Figure 37. SEM image of superior part of filament two from device four, x1000.

5.1.3 Filament 2.8

As it is said before, not all the parts of the filament are analyzed, only those where there is something of interest. In this filament, the only analyzed parts are: Center, center-superior and superior part, assuming it is a symmetrical filament. Just as reminder, the time of this filament is around 183 hours.

The center part of the filament is represented in Figure 38, and it is observed that the corrosion on the surface is notable, furthermore, the erosion produced by the ionic flow during the discharge is also appreciable, even more in the left part of the image.

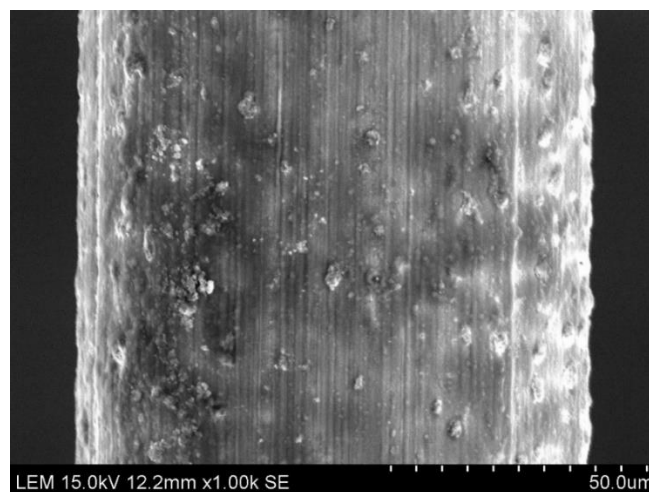


Figure 38. SEM image of center part of filament two from device four, x1000.

The corrosion and erosion produced during corona discharge on the left part of the filament is more remarked in Figure 39 & 40, where we can conclude that during the process, the left part was facing the mass electrode, producing more corrosion in that part than the rest of the surface of the filament.

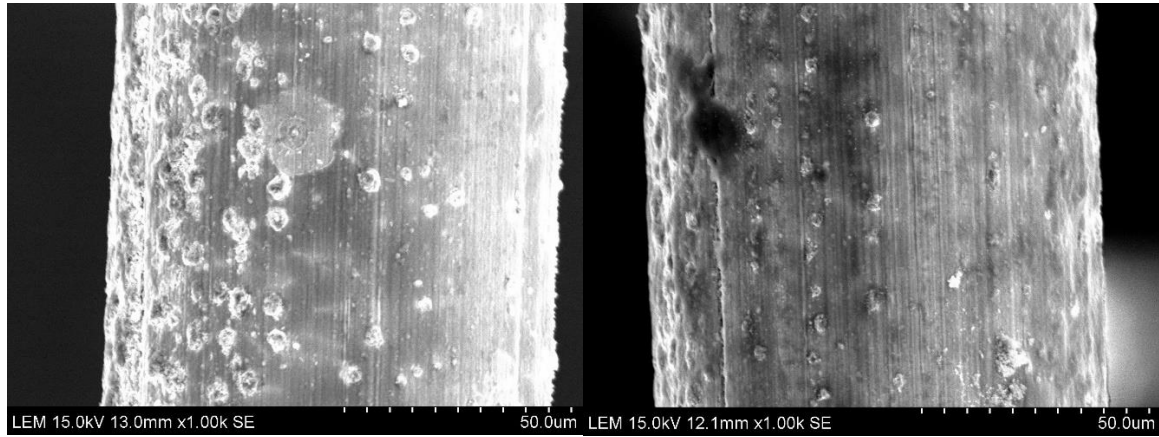


Figure 39. SEM images of center-superior part of filament two from device four, x1000.

As we can observe, the corrosion and erosion of the center-superior part of the filament is more noticeable, especially in the left image, where the tungsten oxide can be clearly observed. Furthermore, the left image of Figure 30 shows a particular morphology that seems to correspond to micro-fusion by arc, that may be produced by arc discharge, previously explained, where a spark is produced between two electrodes discharging them. The arc discharge is a high intensity and high temperature spark that may lead to heated up regions on the surface and is a phenomenon that sometimes occur in these devices, so it is a good hypothesis to consider regarding the micro-fusion morphology. These areas where the micro arches produce craters, appear to be more corroded, which indicates that the arches accelerate the corrosion process.

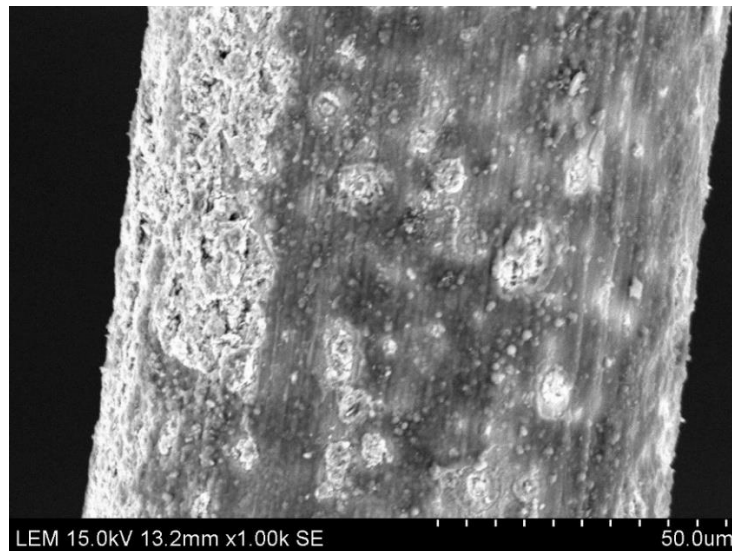


Figure 40. SEM image of superior part of filament two from device four, x1000.

Finally, the superior part of the filament is shown in Figure 40, where it is definitely more corroded than the last two parts of the same filament. The corrosion is more remarkable either in the left region and the right region. The effect of erosion is very considerable in this image, because the left part is much more corroded than the right part.

Moreover, the images of filament 2.8 were subjected to further analysis using ImageJ and the diameters measured with this software were around 100 μm , which is the initial diameter of the filament. A possible explanation of this phenomenon is exposed below.

Finally, the conclusion of these images from different filaments subjected to different times is that corrosion is highly dependent on the time subjected to corona discharge, comparing filaments 2.4 and 2.8, it is clearly seen that filament 2.8 is more corroded than filament 2.4, being their times equal to 183 and 40 hours respectively. An interesting deduction is that less corroded filaments seems to have less diameter than more corroded ones, however, this is considered as a variation on the dimensions as a consequence of manufacturing errors. What is observed in both filaments, is that the part facing the mass electrode undergoes more corrosion. This is because the corrosion process is helped by the erosion induced by corona discharge. Furthermore, the corrosion is also higher at this part because the generated plasma tends to orientate towards the mass electrode, producing an accumulation of ions in that part, increasing the corrosion process.

5.2 Computational results

The results obtained by the simulation performed in COMSOL are divided depending on the filament diameter. At first, one simulation for 25 μm and other for 100 μm are performed at times up to 90 and 180 hours respectively. The results of both simulations and its discussion are exposed following.

5.2.1 Filament 25 μm

The 25 μm diameter filaments are the first to be tested in order to compare it with the broken filament obtained previously. The equation that governs the current density is an approximation obtained by Equation 33 with the values represented in Table 3 below.

Diameter (μm)	25
Voltage (V)	4000
Time (h)	90
Time step (h)	0.5
Current Density (A/m^2)	Equation 33

Table 3. Simulation data for 25 μm filaments.

The simulation time is set to 90 hours because it is the time at which the broken filament from previous section was tested, with a time step of 0.5 hours, leading to 180 steps in total. In addition, the voltage is 4 kV, the same for every 25 μm filament.

The time dependent with initialization study is ran and 50974 degrees of freedom are solved in the simulation. This number is higher than 100 μm filaments because the number of surface elements is maintained the same despite the diameter is smaller, therefore, the total nodes is higher. After 4 minutes and 22 seconds of simulation, the results are finally obtained.

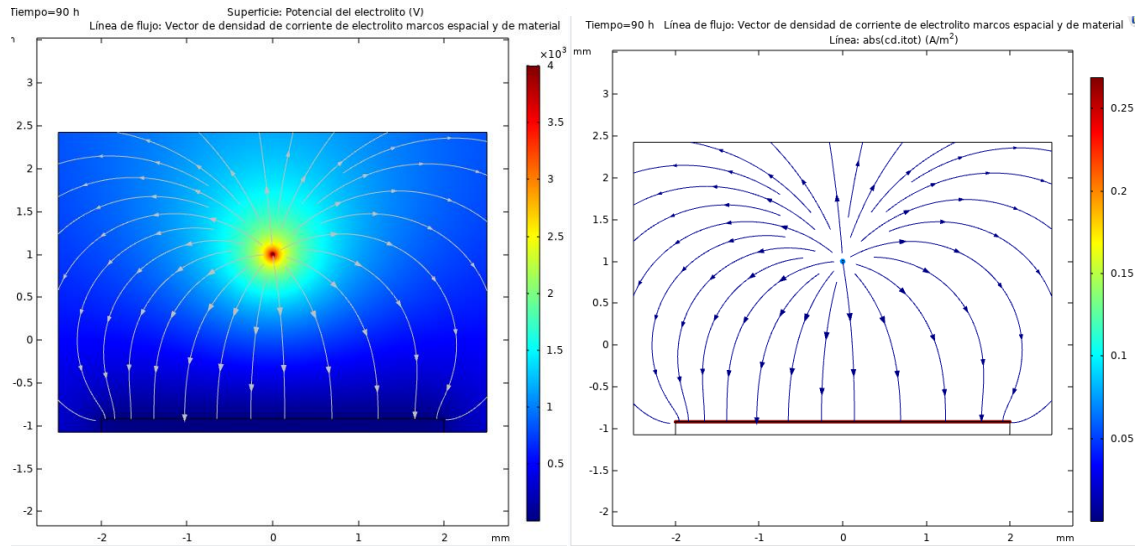


Figure 41. Voltage map (left) and current density vector graph (right) of 25 μm filament generated by COMSOL Multiphysics.

As it is observed, the voltage on the corona electrode reaches the highest value, specifically 4 kV, whereas the voltage value on the mass electrode is 0 V. Therefore, the vector lines go from the corona electrode towards the mass electrode, expressing the potential difference between them. On the right image, it is observed that, although the lines go from corona to mass, the current density on the surface of mass electrode is higher than the current density on the corona electrode, this is because the mass electrode undergoes a more aggressive corrosion, leading to this higher value.

Following, the thickness variation of both electrodes is represented in Figure 42 in μm . Despite it is difficult to quantify from this image, it is clearly seen that the mass electrode has a higher thickness variation than the corona electrode.

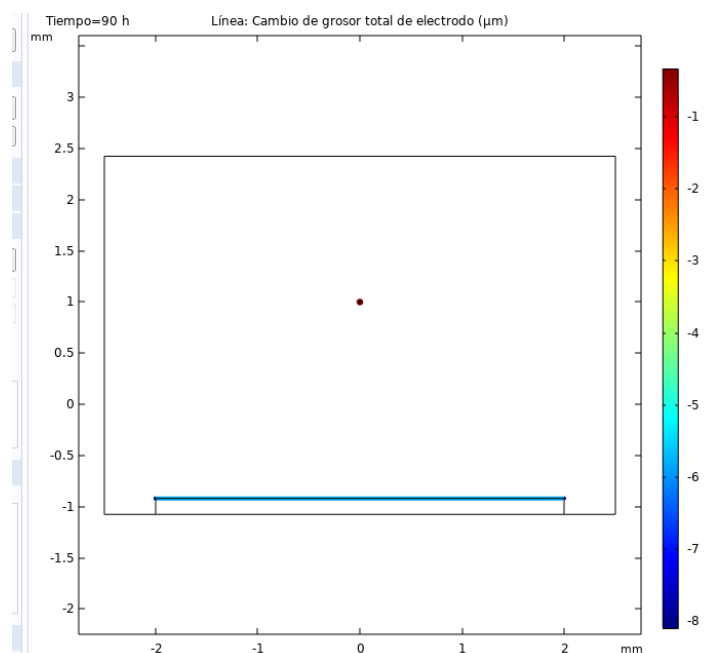


Figure 42. Thickness variation map of 25 μm filament generated by COMSOL Multiphysics.

In order to more visible comparison between corona and mass electrode thickness variation, the data of both electrodes are exported from COMSOL in a text file and, subsequently, converted into excel with the aim of plotting a graph containing both variations. To do so, a 1D graph group is created in COMSOL results, following a linear graph is created in the 1D graph group, and the surface of corona electrode is selected. The next step is to define the electrode thickness variation in the y-axis and time in the x-axis and export the data. The same process is done for mass electrode.

In Figure 43 the plots obtained directly by COMSOL are presented. In order to make the comparison easier to observe, both graphs are going to be merged in one using excel, as it is said before. In the case of the mass electrode, the tendency line of the graph is going to be used because that is the real thickness variation value simulated by the program.

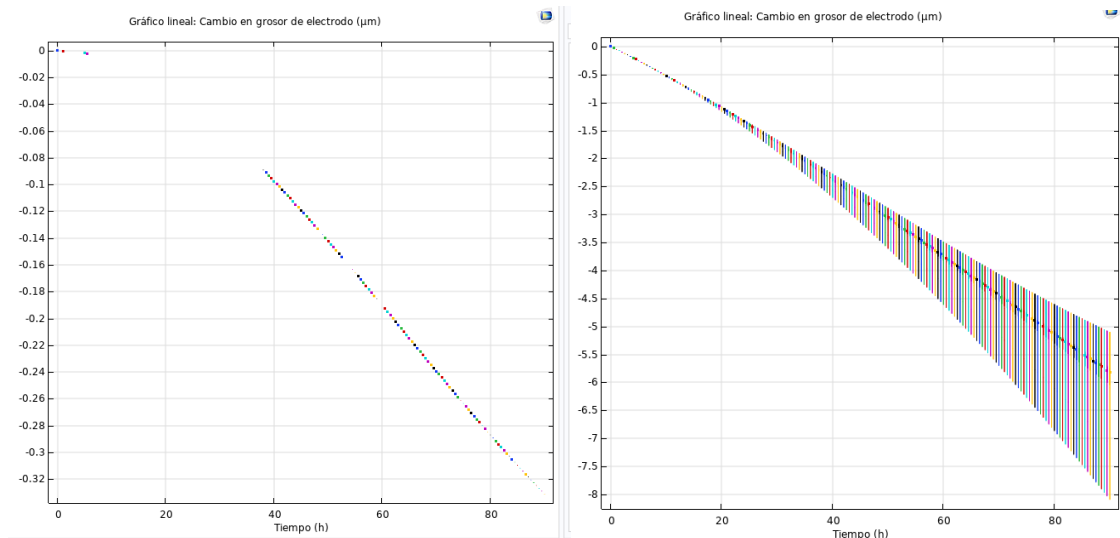


Figure 43. Thickness variation graphs corona electrode (left) and mass electrode (right) of 25 μm filament by COMSOL Multiphysics.

In Figure 44, the comparison between the thickness variation of mass and corona electrodes is presented in one graph.

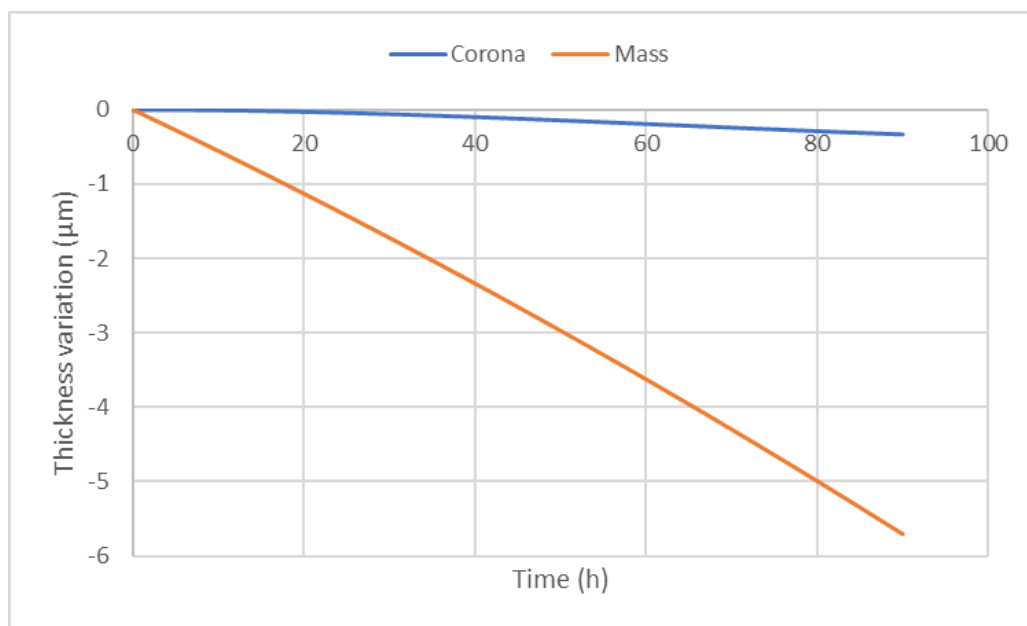


Figure 44. Thickness variation comparison between corona and mass electrode. 25 μm generated by COMSOL Multiphysics.

As it is observed, the corona electrode degrades at a less rate than the mass electrode. Specifically, the corona electrode corrodes 0.33 μm after 90 hours of corona discharge, which is pretty low. Mass electrode corrodes 5.7 μm after the same time of exposure. The high difference between both electrodes can be explained by the ion flux moving toward mass electrode. The active ion species and the constant movement of the air produces this

increase in the degradation rate of the aluminum. Experimentally, it was observed that the corrosion rate stabilizes at times higher than 100 hours for 25 μm filaments.

5.2.2 Filament 100 μm

As it is stated in the previous chapter, the equations governing the current density are approximations obtained experimentally by CEDRION. A time dependent study with initialization has been performed to 180 hours with an hour steps for this filament, as it is represented in Table 4.

Diameter (μm)	100
Voltage (V)	5000
Time (h)	180
Time step (h)	1
Current Density (A/m^2)	Equation 32

Table 4. Simulation data for 100 μm filaments.

First of all, the voltage distribution through the region is represented in Figure 41, where the closest area to the corona electrode have higher potential, close to 5 kV, and the mass electrode has the least potential, generating a big potential difference in the region. It is also observed the flux of electrolyte current density that flows from corona electrode to mass electrode because of the potential difference.

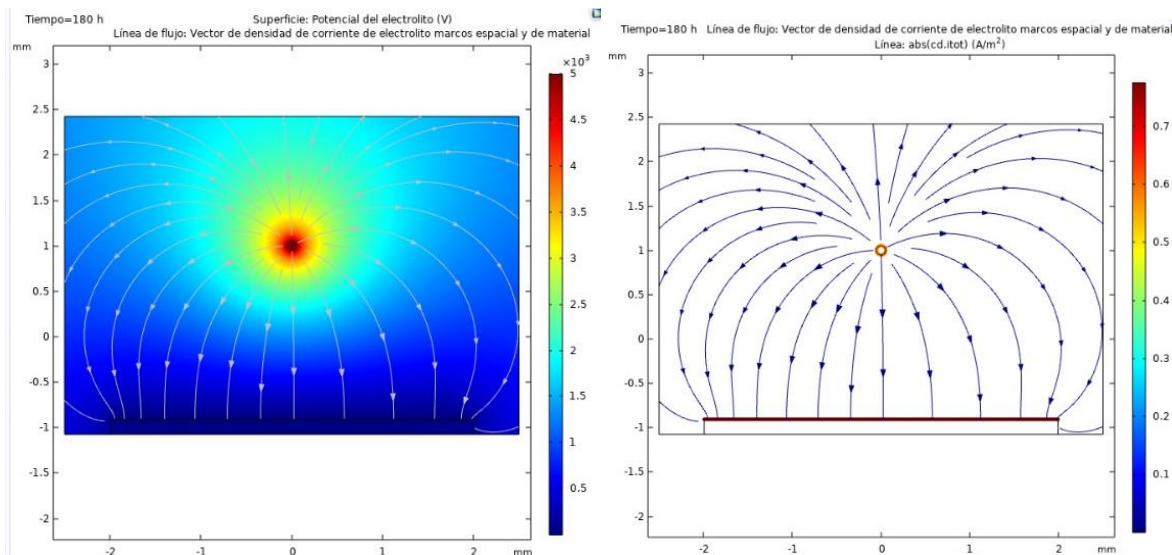


Figure 45. Voltage map (left) and current density vector graph (right) of 100 μm filament generated by COMSOL Multiphysics.

Moreover, in the right image of Figure 45, it is observed that the current density is higher right on the surface of the mass electrode than the surface of the corona electrode, this is because of the mass electrode is corroded at a higher rate than corona electrode due to the ion flux of reactive species and the effect of the electric field generated by corona electrode.

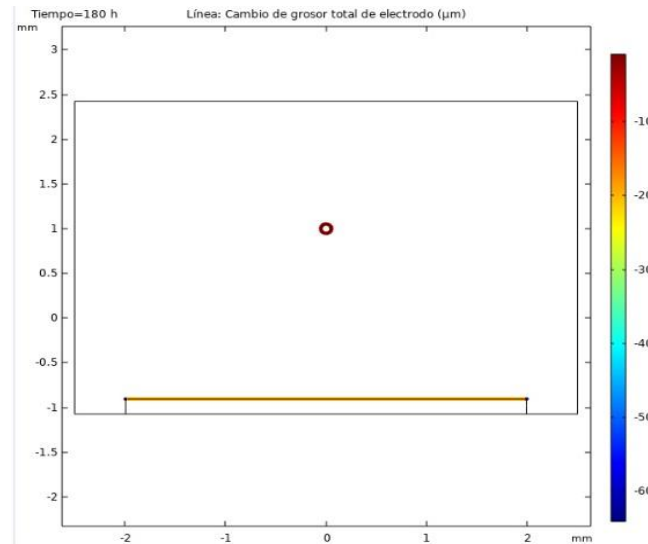


Figure 46. Thickness variation map of 100 μm filament generated by COMSOL Multiphysics.

Following, in Figure 46, the thickness variation of both electrodes as a result of the calculation done by the program is shown. It is concluded from this image that the thickness variation of mass electrode is higher than corona electrode after 180 hours, hence, the initial corrosion is also higher. In the following image, the graphs generated by the program corresponding to thickness variation of both electrodes are shown.

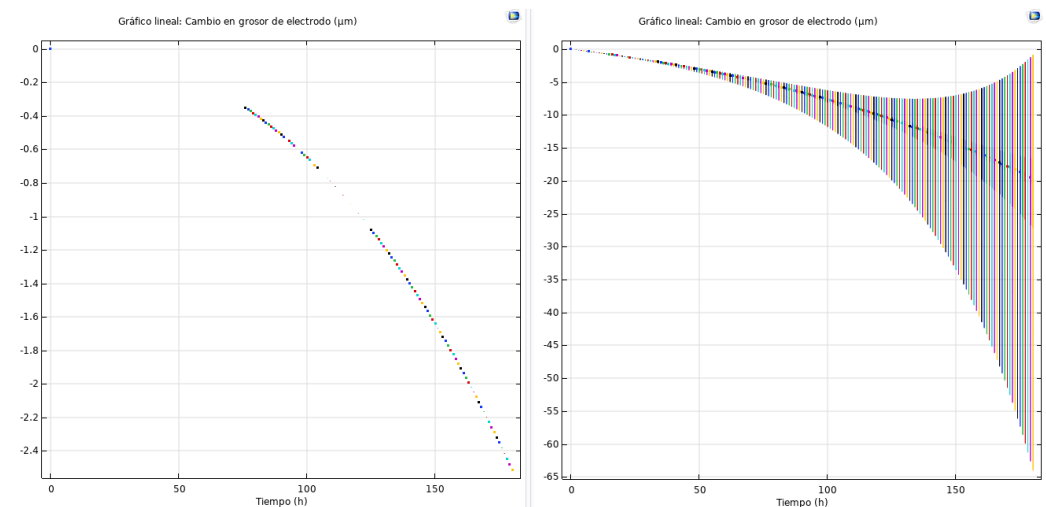


Figure 47. Thickness variation graphs corona electrode (left) and mass electrode (right) of 25 μm filament by COMSOL Multiphysics.

In order to make a noticeable comparison for further analysis, the thickness variations of both electrodes are exported as a text file, and then, converted and plotted in excel. In Figure 48 a comparison between thickness variation of corona and mass electrodes is shown.

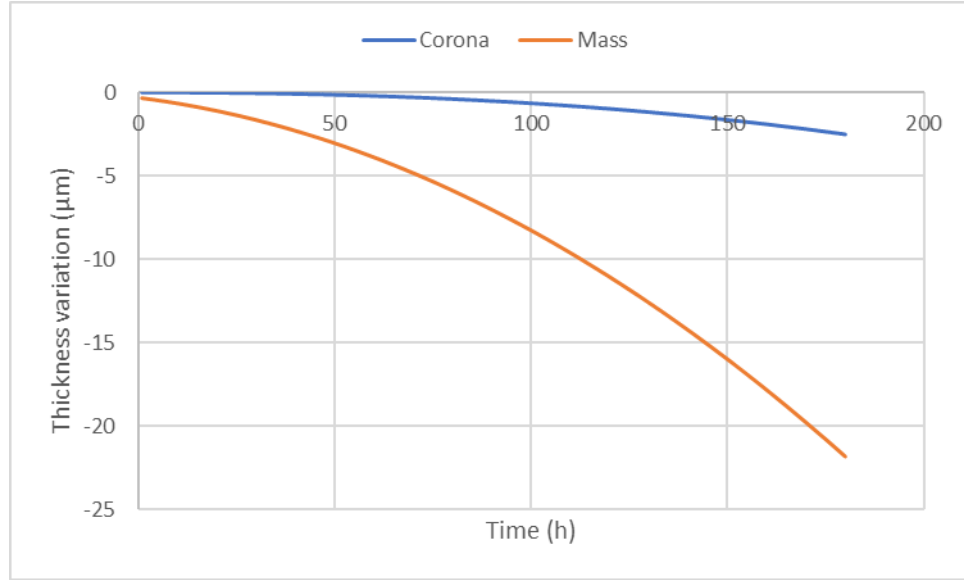


Figure 48. Thickness variation comparison between corona and mass electrode. 100 μm generated by COMSOL Multiphysics.

The variation of thickness decrease rate between corona and mass electrode is very remarkable, being mass thickness decrease rate up to 4 times higher than corona thickness decrease rate, at 180 hours. Being 2.5 μm of the corona electrode converted into tungsten oxide while there are around 21 μm of mass electrode converted into alumina.

As a conclusion, as long as the time keeps increasing in the range of 0 to 200 hours, the thickness decrease difference between both electrodes will keep increasing being the mass electrode heavier than corona. After 200 hours, experimentally has been discovered that the corrosion is stabilized, this means that the rate is maintained in a small range.

5.2.3 Erosion rate calculation

Apart from the corrosion simulation, erosion rate is also quite important in corona discharge processes, as it is said before. CEDRION determined experimentally the erosion rate of tungsten from results obtained and applying the following equation^[28]:

$$S_k = \frac{A_D}{Q_k} \left(1 - \frac{R_0}{R_k} \right) \quad (34)$$

With S_k being the erosion rate, R_k and R_0 the final and initial values of resistance per unit length, respectively, A_D is related to the area of corona electrode and Q_k is the charge transferred per unit of length.

The erosion rate in a 100 μm and a 25 μm filament are represented in Figure 49. The substantial difference in the x-axis is because the values obtained from the 25 μm filaments are up to 138 hours of testing, whereas the 100 μm filaments are up to 380 hours.

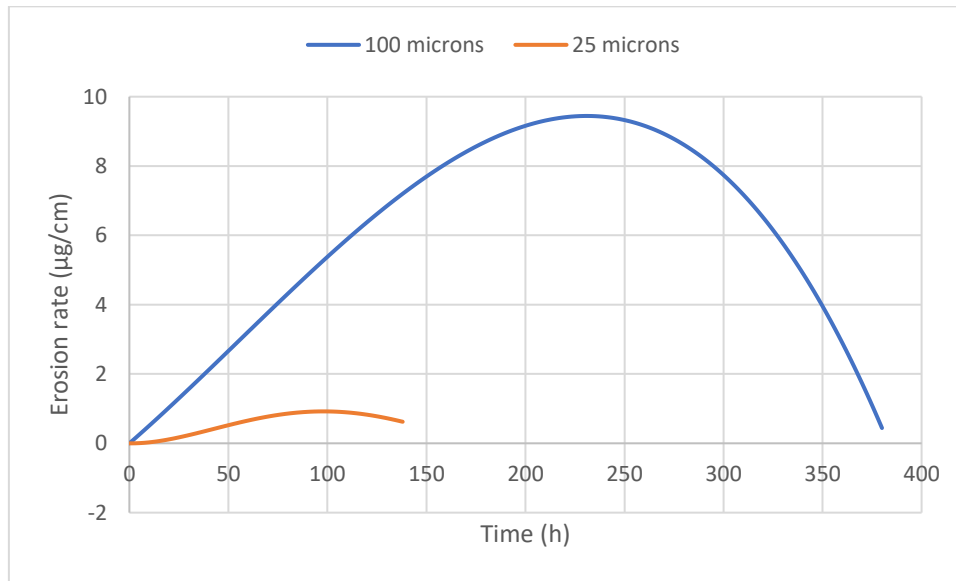


Figure 49. Erosion rate of a 100 μm filament.

In Figure 49 it is appreciated that the erosion rate of 100 μm reaches a maximum value between 200 and 250 hours of testing. While the 25 μm reaches a maximum value around 100 hours of testing. The erosion rate of 25 μm filaments is considerably lower than 100 μm filaments, this may be produced by the smaller surface of the 25 μm filaments compared to the 100 μm .

5.3 Discussion

In this part of the document, the experimental and computational results previously obtained are compared in order to make relevant conclusions of the degradation of electrodes subjected to corona discharge processes.

First of all, the 25 μm filament computational results are going to be compared to the SEM results. The 25 μm filament observed in SEM is broken after 90 hours of testing.

Comparing with the results obtained in the simulation, the failure of the filament does not fit with the results, because at 90 hours the tungsten thickness that has been oxidized does not reach 1 μm . As it is said before, the erosion rate of a 25 μm reaches a maximum around 100 hours, but it is still a low erosion rate. In conclusion, the failure of this filament is probably not produced directly by corona discharge, but due to manufacture errors or excessive applied stresses during device set-up.

The comparison of 100 μm filaments computational results with SEM results seems to fit very well. As it is observed in Figures 35 to 37, the corrosion produced at 40 hours is not very remarkable, which fits with the graph shown in Figure 48 because the corrosion is increased with time, being barely appreciated at 40 or less hours and being much more noticeable at high time values, like 180 hours. This is corroborated by the images represented in Figures 38 to 40 where the corrosion observed in the filament is very pronounced comparing with the images at 40 hours. Furthermore, the erosion is also appreciable on the left part of the filament, that faces the mass electrode and, therefore, where the erosion is located. In addition, the erosion rate at 180 hours is relatively high, hence, it is more visible on the filament surface.

Following, in order to discuss the corrosion undergone by filaments with different diameters, the thickness variation graphs are merged and represented in Figures 50 & 52. Prior to start the discussion, it has to be detailed that the 100 and 25 μm filaments and its respective voltages, are designed to have the same or at least similar power. Hence, comparing the graphs is very interesting in order to find the least susceptible to corrosion.

As it is observed in graph 50, the thickness variations are almost the same at 60 hours, point where the graphs diverge and the 100 μm filament increases more the thickness variation than 25 μm . In order to observe a more detailed comparison between them, the values of thickness are going to be divided by its radius, with the aim of obtaining a more realistic and comparable graph.

In Figure 51, the relative thickness variation respect to the radius of the filament is presented. It is deduced from this graph that the corrosion undergone by 25 μm corona is, respect to its radius, higher than the 100 μm , reaching to its double value and higher. Furthermore, the thickness variation is stabilized first in 25 μm while 100 μm keeps increasing through time. Hence, watching this, the corrosion is higher on 25 μm , at least within the range 0 to 100 hours.

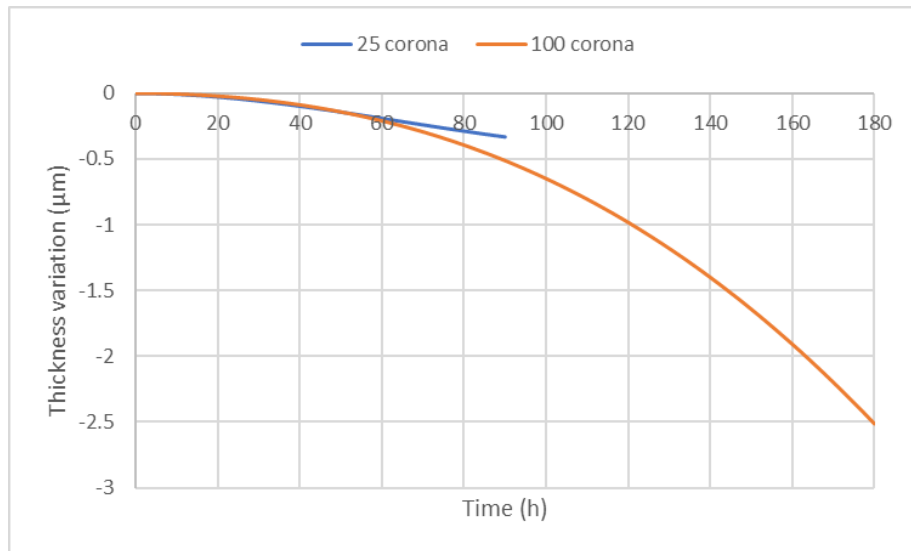


Figure 50. Corona electrode thickness variation comparison between 25 and 100 μm .

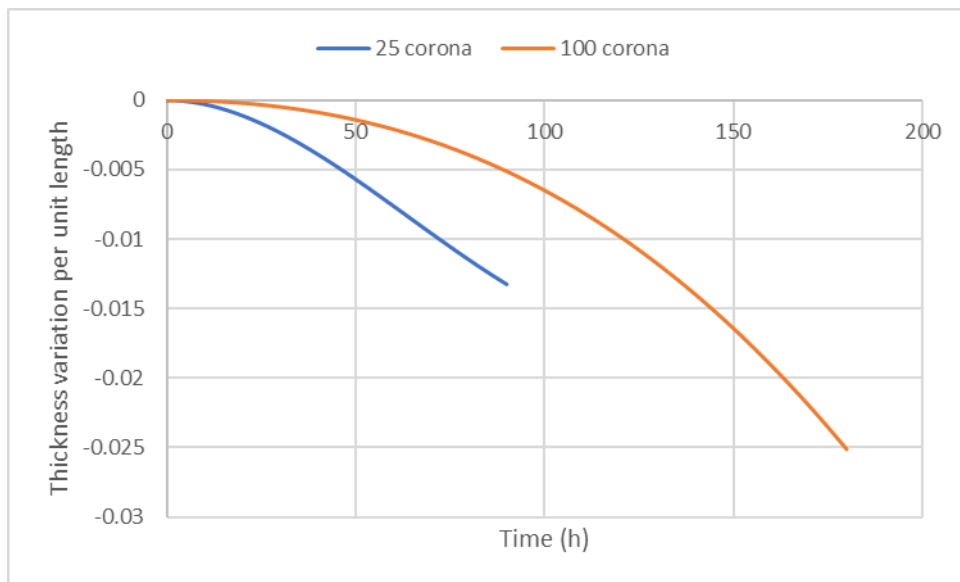


Figure 51. Corona electrode thickness variation per unit length comparison between 25 and 100 μm .

Finally, the comparison between the mass electrodes used in 25 and 100 μm corona electrodes testing is represented in Figure 52. It is observed that the thickness variation is similar up to 50 hours and after that point the mass electrode of 100 μm filament degrades faster than mass electrode of 25 μm filament.

To conclude, it is observed that the 25 μm filaments degrades faster than the 100 μm filaments, however, the mass electrode of 25 μm filaments degrades slower than the mass electrode of 100 μm filaments. This higher degradation results from the electric field increase on the surface of the electrode, which lead to a higher ionization and, hence, to a higher corrosion.

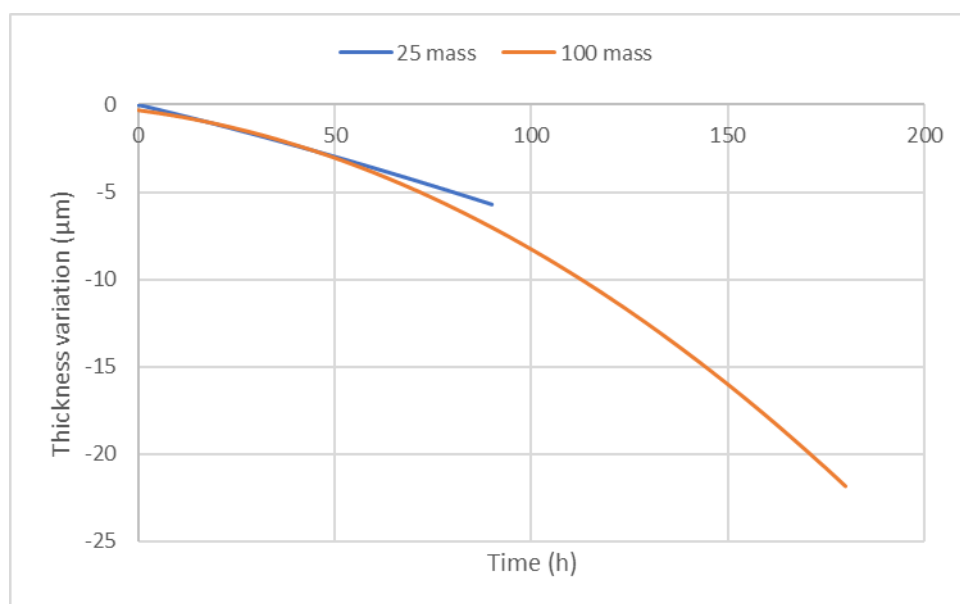


Figure 52. Mass electrode thickness variation comparison between 25 and 100 μm corona electrodes.

6 Conclusions

In this chapter the more relevant conclusions are collected taking into account the limitations of this work.

- The first conclusions obtained from this project is that the corona electrode gets more corroded on the part of the surface that is facing the mass electrode during the corona discharge process. This is corroborated by the images obtained by SEM and the erosion process that takes places during the discharge. Furthermore, the failure mechanism of filament 1.5 is not directly induced by corona discharge, hence, the failure mechanism is related to manufacture errors or residual stresses.
- The second point is that the corrosion undergone by the materials during corona discharge is not linear. It starts at a low rate and increases with time as observed in the simulations and the filaments 2.4 and 2.8, both with 100 μm diameter.
- The third point is that when the diameter of the corona electrode is decreased, the power generated increases. It is theoretically demonstrated in chapter two and deduced from results, but the corrosion undergone by those filaments is reduced, especially after 60 hours of testing, as observed in Figure 50. However, the corrosion of the filament compared with the filament size increases from the beginning, as shown in Figure 51, so it is concluded that the degradation for a certain filament diameter is increased when its diameter is reduced. This is produced as a result of the electric field increment on the filament surface, where the ionization occurs, leading to a higher ionization and, therefore, a higher

corrosion of the electrode. This is demonstrated at times under 100 hours, above that time it cannot be confirmed.

- Finally, it is also proved that the mass electrode degrades faster than the corona electrode because the tungsten is a refractory material and aluminum is more active. In addition, the active ion species that flow toward the mass electrode increases its degradation. Furthermore, the mass electrode performing with a 100 μm filament undergoes a higher corrosion than with a 25 μm , especially above 50 hours of testing, this is observed in Figure 52.

7 Future lines

The knowledge of the corrosion process involved in corona discharge needs to know the products formed due to metal corrosion. The small dimensions of filaments cause EDX to be insufficiently accurate. The following points would be necessary:

1. Study “*as received*” filaments in order to have a reference to compare the different filaments after different condition and times service.
2. Analyse the products of corrosion to approximate the corrosion equations in the mathematical model. It is necessary to perform analyses with more accurately than EDX, such as XRF.
3. Test different filaments under conditions derived of previous simulations.
4. Perform a more optimized 2D simulation a well as a 3D simulation model.
5. Calibration and optimization of the mathematical model with tests results.

References

- [1] Cogollo, Mar, Pedro M. Balsalobre, Andrés Díaz Lantada, and Héctor Puago. 2020. *“Design and Experimental Evaluation of Innovative Wire-to-Plane Configuration For Atmosphere Corona Discharge Cooling Devices.”*
- [2] Morvová, Marcela. 1998. *“Dc corona discharges in CO₂–air and CO-air mixtures for various electrode materials.”*
- [3] Wang, Lawrence K., Norman C. Pereira, and Yung-Tse Hung. 2004. *“Air Pollution Control Engineering.”* Totowa, New Jersey: Humana Press.
- [4] Chang, Jen-Shih, Phil A. Lawless, and Toshiaki Yamamoto. 1991. *“Corona Discharge Processes.”*
- [5] Wilson, Jack, Hugh D. Perkins, and William K. Thompson. 2009. *“An Investigation of Ionic Wind Propulsion.”*
- [6] Kim, Hyun-Ha. *“Nonthermal plasma processing for Air-pollution control: A historical review, Current issues and Future prospects.”*
- [7] Goldman, M., A. Goldman, and R. S. Sigmond. *“The corona discharge, its properties and specific uses.”*
- [8] Jewell-Larsen, Nels E. *“Electrohydrodynamic movers for thermal management.”*
- [9] Horwitz, Mara E. M., S. Gray Horwitz, and Chris M. Horwitz. *“Corona Emission and Ozone Production by Carbonized and Oxidized High-Voltage Wires.”*
- [10] Boulloud, A., and J. Charrier. *“Positive glow corona at a point in air: Stabilization of the dual-patch regime.”*
- [11] Conesa, Antonio J., Mario Sánchez, Marina León, and Ángel Cabrera. *“Some geometrical and electrical aspects on the wire-to-cylinder corona discharge.”*
- [12] Lassner, Erik, and Wolf-Dieter Schubert. *“Important aspects of tungsten chemistry.”*
- [13] Utembe, S.R., G.M. Hansford, M.G. Sanderson, R.A. Freshwater, K.F.E. Pratt, D.E. Williams, R.A. Cox, and R.L. Jones. *“An ozone monitoring instrument based on the tungsten trioxide semiconductor.”*

- [14] Williams, David E., and Keith F.E. Pratt. *“Microstructure effects on the response of gas-sensitive resistors based on semiconducting oxides.”*
- [15] Lukes, P., M. Clupek, V. Babicky, and P. Sunka. *“Erosion of needle electrodes in pulsed corona discharge in water.”*
- [16] Lukes, Petr, Martin Clupek, Vaclav Babicky, Irena Sisrova, and Vaclav Janda. *“The catalytic role of tungsten electrode material in the plasmachemical activity of a pulsed corona discharge in water.”*
- [17] Materiales Estructurales II: Seleccin de materiales, aleaciones de aluminio.
- [18] Davis, Joseph R. *“Corrosion of aluminum and aluminum alloys.”*
- [19] Materiales Metálicos III: Superalloys base Níquel.
- [20] Shi, Chaoao. *“Numerical studies of electrohydrodynamic flow induced by corona and dielectric barrier discharges.”*
- [21] Cain, Andy, and David Kan. *“Corrosion modeling: Mechanisms, challenges and numerical solutions.”*
- [22] Kashkoush, Ismail, Ahmed Busnaina, Frederick Kern Jr., and Robert Kunes. *“Ultrasonic cleaning of surfaces: An overview.”*
- [23] Reimer, L. *“Scanning electron microscopy: physics of image formation and microanalysis.”*
- [24] Shindo, Daisuke, and Tetsuo Oikawa. *“Energy dispersive X-ray spectroscopy.”*
- [25] Simon, Andrew H. *“Handbook of the thin film deposition: Sputter processing.”*
- [26] Rasband, Wayne. *“ImageJ, an introduction.”*
- [27] Olegovna, Galina, Nataliia Beloshitskaya, Viktor Sychenko, and Vitalii Liashuk. *“The effect of ozone on wet atmospheric corrosion of aluminium of high-voltage lines.”*
- [28] Ismalov, Rafael Sh., and Yuriy A. Krishtafovich. *“Erosion and Lifetime of Tungsten, Gold, and nichrome wire anodes in an ultrac corona in air.”*